Evaluating the efficiency and water recovery of direct contact membrane distillation (DCMD) systems with dual heat recovery and recirculation of retentate and permeate

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Abstract

A transient model for hollow-fiber direct contact membrane distillation (DCMD) modules with dual heat recovery and partial recirculation of the retentate stream is employed to evaluate a desalination plant in six modes of operation—batch, semi-batch, volume-control, salinity-control, continuous, and steady state—focusing on applications in sites under hydrological stress, such as the semi-arid Brazilian north-eastern region. Adding a second heat exchanger (HX-2) that recovers thermal energy from the hot retentate before its rejection raises the averaged gain output ratio in batch mode from $\overline{\mathrm{GOR}} \approx 0.56$ (no heat recovery) to $\overline{\mathrm{GOR}} \approx 1.82$ (with both HX-1 and HX-2 active), while lowering the averaged specific thermal energy consumption from $\overline{\mathrm{SEC}}_{th} \approx 1.24$ to ≈ 0.39 kWh/L. This gain originates from the sensible enthalpy still carried by the retentate stream at the module outlet, which would otherwise be discarded together with the concentrated brine, and is most pronounced under feed-limited (FLR) conditions with the mass-flow-rate ratio α close to its critical value α^*, since only in this regime does the distillate stream possess sufficient heat-capacity flow rate to sustain the maximum attainable evaporation of the feed. Over an 8-h simulation, batch operation yields the lowest specific thermal energy consumption

($\overline{\mathrm{SEC}}_{th} \approx 0.39$–$0.43$ kWh/L) because the tank is emptied and refilled sequentially, keeping the mean feed salinity—and thus the transmembrane vapor-pressure driving force—favorable for most of each cycle. Semi-batch operation, in contrast, delivers the highest water recovery ratio (RR $\approx$ 76–92%, against only 2–5% in the pure steady-state mode without recirculation), since fresh brackish water is continuously supplied at the same rate at which distillate is produced and no volume is wasted during flush-and-refill transients; the volume-control mode delivers an intermediate compromise (RR $\approx 76\%$ at $\overline{\mathrm{SEC}}_{th} \approx 0.40$ kWh/L for $C_{s,\max} = 120$ $\mathrm{g}_s/\mathrm{L}_{mix}$). Hence, if the primary interest is minimizing thermal energy cost, batch is the best option; in regions subject to seasonal hydrological stress, the semi-batch mode is preferable, provided that the operational costs of pumps and control devices are subsequently included in a broader thermoeconomic analysis.

Keywords: Desalination; DCMD system; heat recovery; retentate recirculation; energy efficiency.

Nomenclature

Latin Symbols (a–m)		Latin Symbols (n–z)	
Symbol	Description	Symbol	Description
a	Specific surface area	N	Molar flux
c_p	Specific heat at constant pressure	N_{fbr}	Number of fibers in the vessel
C	Thermal capacity	Nu	Nusselt number
C_s	Salt concentration	p	Pressure
d	Diameter	Pr	Prandtl number
D	Diffusivity / diffusion coefficient	q_c	Heat transferred to the cold sink
f	Friction factor	q_h	Heat transferred from the hot source
h	Convective heat transfer coefficient	q_{HX}	Heat transferred in the heat exchanger
h_{fg}	Enthalpy / heat of vaporization	Q	Rate of heat transfer
j_{wl}	Distillate flow rate	r	Radius
k	Thermal conductivity	R	Universal gas constant
L	Hollow-fiber module length	Re	Reynolds number
m	Mass	T	Temperature
$\dot{m}$	Mass flow rate	V	Volume
M	Molar mass	$\dot{V}$	Volumetric flow rate
		u	Stream velocity
		z	Position along the module
Greek Symbols		Subscripts / Superscripts	
Symbol	Description	Symbol	Description
α	Membrane effectiveness	a	Relative to air

Nomenclature (continued)

Acronyms			
Symbol	Description	Symbol	Description
AGMD	Air Gap Membrane Distillation	PGMD	Permeate Gap Membrane Distillation
CGMD	Conductive Gap Membrane Distillation	RO	Reverse Osmosis
DCMD	Direct Contact Membrane Distillation	RR	Recovery Ratio
DLR	Distillate Limiting Ratio	SEC	Specific Energy Consumption
GOR	Gained Output Ratio	SGMD	Sweeping Gas Membrane Distillation
FLR	Feed Limiting Ratio	V-AGMD	Vacuum Air Gap Membrane Distillation
HICO	Hot In, Cold Out	VMD	Vacuum Membrane Distillation
HOCI	Hot Out, Cold In		
HX	Heat Exchanger		
MD	Membrane Distillation		

1 Introduction

Freshwater scarcity, mainly due to population growth, has attracted the attention of researchers and industry for the development and improvement of techniques for the treatment, purification, and desalination of salt water. Among the main desalination techniques, those involving thermal separation methods, such as membrane distillation, require more energy due to the need to vaporize the water since the water vaporization enthalpy is about 2,400 kJ/kg [Sharqawy et al., 2010], while Gibbs free energy for seawater, which governs the energy use of the reverse osmosis method, is 2.7 kJ/kg [Elimelech and Phillip, 2011]. Therefore, one of the challenges that must be overcome for the application of thermal methods such as membrane distillation (MD) in water desalination, which aims to be competitive against the conventionally employed reverse osmosis (RO) process in small and medium-sized systems (< 100 m^3/day of water produced), is to minimize the demand for thermal energy, improving thermal efficiency.

The membrane distillation process enables the use of residual heat from other thermal processes or the heat available from renewable sources, such as solar collectors. However, heat losses by conduction through the membrane further increase the thermal consumption, which, when added to high-water consumption, limits the large-scale use of these systems [Khayet and Matsuura, 2011]. Nevertheless, according to Guan et al. [2015], when compared to conventional desalination processes, such as multi-stage flash distillation, multi-effect distillation, and reverse osmosis, the MD process has several benefits: low salinity sensitivity and high salt rejection (100% content, theoretically), low membrane vulnerability to fouling, greater performance under low severity operating conditions, and the viability of using low-grade waste heat from other thermal processes or renewable sources, such as solar collectors, as previously mentioned [Curcio and Drioli, 2005, Charcosset, 2009, Drioli et al., 2015].

Concerning its technology, according to the method of obtaining the vapor pressure gradient through the membrane, the MD is classified in the following configurations [Zhang et al., 2015, Swaminathan et al., 2018]: DCMD (Direct Contact Membrane Distillation), AGMD (Air Gap Membrane Distillation), SGMD (Sweeping Gas Membrane Distillation), VMD (Vacuum Membrane Distillation), PGMD

(Permeate Gap Membrane Distillation), CGMD (Conductive Gap Membrane Distillation), V-AGMD (Vacuum-Enhanced Air Gap Membrane Distillation), among other variants. Each configuration has its advantages and intrinsic disadvantages, so the choice of one of them is still a research theme, which is usually directed through metrics such as GOR (Gained Output Ratio) and SEC (Specific Energy Consumption), which are parameters for energy efficiency analysis. The DCMD is the most studied and straightforward configuration among them, widely used for desalination studies and concentration of aqueous solutions.

In general, the increase in the thermal efficiency of MD systems is achieved by improving the performance of membranes (by choosing membranes with low thermal conductivity, adequate porosity, pore size and pore size distribution, and thickness), as well as by reducing the amount of external heat to be supplied to the process [Gostoli and Sarti, 1998, Khayet et al., 2006, Al-Obaidani et al., 2008, Yao et al., 2013, Li et al., 2014, Swaminathan et al., 2018].

The heat supplied to the process can then be significantly reduced by the reuse of the heat available in the output currents of the desalination module through external heat exchangers, extending the applicability of the MD process and improving energy efficiency [Gilron et al., 2007, Lee et al., 2011, Hausmann et al., 2012, Lu and Chen, 2012, Zhang et al., 2015, Gonzalez-Bravo et al., 2015, Kim et al., 2015, Maheswari et al., 2015, Lin et al., 2014, **?**, Christie et al., 2020, Lisboa et al., 2021, Sampaio, 2022]. Obviously, operating conditions such as temperature, the feed and distillate flow rates, salinity of the feed stream, and its recirculation, among others, significantly affect the MD process and can improve or reduce process efficiency [Khayet et al., 2000, Wang et al., 2009, Zhang et al., 2010, Fan and Peng, 2012, Alkhudhiri et al., 2012, Guan et al., 2014].

Additionally, since MD presents a low recovery ratio per pass (ratio between the production rate of pure water and the feed flow), using some form of recirculation of the brine/retentate stream is mandatory to increase efficiency. In this way, by considering a CGMD module, Swaminathan and Lienhard [2018] describe four operation strategies that enable high global recovery using a single-pass low recovery process (batch recirculation, semi-batch recirculation, continuous recirculation, and continuous multi-stage recirculation). The first two operations are classified as tran-

sient/discontinuous processes, while the last two are steady state ones. According to these authors, who studied the salinity variation from a lower limit of 70 g/kg to an upper limit of 260 g/kg, the batch operation is the best option since GOR is about twice for the same distillate flow when compared to the continuous recirculation operation. In their analyses, using a tank with a high volume to minimize salinity variation, they adopted a quasi-permanent approach, using a steady state MD model for instantaneous performance evaluation so that transient effects between each cycle could be ignored. In this respect, cycling time is thus considered to be very small.

Sampaio [2022] modeled a seawater desalination DCMD plant with heat recovery of the recirculating hot distillate stream by using a DCMD module composed by a bundle of hollow-fiber tubes. He combined analytical solutions for the mass transfer problem across the membranes with the finite volume method for solution the steady-state heat transfer problem. Still, based on Christie et al. [2020] and Lin et al. [2014], and to extend their works, Sampaio [2022] developed an approximation for the optimal value of the ratio between permeate and feed flow rates, and for the maximum mass recovery rate that, differently from the expressions developed by Christie et al. [2020] and Lin et al. [2014], take into consideration the membrane thermal efficiency.

Thus, based on the works of Swaminathan et al. [2018], Christie et al. [2020], Lisboa et al. [2021], and Sampaio [2022] and extending their analyses, the present work aims at the development of physical-mathematical modeling, in the unsteady-state regime, and subsequent numerical simulation of a hollow-fiber DCMD system with double recirculation and heat recovery. The main goal is firstly to evaluate the potential of heat recovery from the retentate stream to increase the energy efficiency of the system to produce distilled water. Furthermore, an application of the model to study the desalination process in regions with high seasonal hydrological stress, such as the semi-arid region of the Brazilian northeastern. In addition to the classical batch, semi-batch, continuous, and steady-state modes of operation, two intermediary modes, namely, mixing tank volume controlling and mixing tank salt concentration controlling, are also introduced. The aim is to verify their effect on the behavior of the system efficiency and on the water recovery ratio (for minimizing the well brackish water consumption). These controlling schemes can also be operated

in conjunction with total or partial recirculation, i.e., when the brine is rejected from the system with fraction rejections equal or different to the unity, respectively.

Table 1 presents, in a condensed way, a brief bibliographic review of the main experimental and simulation works on the thermal efficiency of MD systems, highlighting the main contributions of each of the works, especially, but not limited to, those works that adopt heat recovery and recirculation of flow streams using hollow fiber membrane configuration.

2 System Description

The MD desalination plant modelled in the present work is characterized by a water desalination system that operates in the transient regime, containing a DCMD module (a vessel with a bundle of hollow fiber membranes), two tanks (a mixing and collector), and running with dual recirculation and dual heat recovery, where the heat available at the output of the module from the feed and distillate streams are recovered through the heat exchangers, HX-1 and HX-2, respectively, as shown in Figure 1. The heat source and sink are constant-temperature heat reservoirs, so the feed and distillate streams reach the specified values of the source/sink temperatures by supplying/extracting exactly the corresponding amount of heat required. This assumption is adopted for system evaluation purposes only since, in practice, with the use of heat exchangers, temperatures are not kept constant, but rather come from sensible waste heat source.

At the initial time, the feed circuit is completely filled with brackish water at the well salinity (c_s^{well}) and ambient temperature (T_{amb}). The distillate circuit is filled with distilled water at the same ambient temperature. The feed stream leaves the mixing tank at state 3 (c_{s3}, $V_3 = V_{feed}$, T_3) and enters HX-1, which heats it to state 4 (c_{s4}, V_4, T_4) by receiving energy from the distillate leaving the DCMD module at state 11 (c_{s11}, V_{11}, T_{11}). The mandatory use of this heat exchanger (HX-1) greatly increases the efficiency of the MD system by reducing the amount of heat (q_h) that must be supplied by the heater to raise the feed stream temperature to the working temperature at state 5 (c_{s5}, V_5, $T_5 = T_{feed}$).

The concentrated module effluent (the retentate) at state 6 (c_{s6}, V_6, T_6) can be

Table 1: Literature review on thermal efficiency in MD systems by heat recovery.

Reference	Heat Recovery			Observations
	Distillate	Distillate	Retentate	
Fane *et al.* (1987)	X	X	X	Experimental/Simulation – Hollow fiber – DCMD
Schneider *et al.* (1988)	X	X	X	Experimental – Hollow Fiber – DCMD
Kurokawa and Sawa (1996)	X	X	X	Simulation – Flat fiber – DCMD
Li and Sirkar (2004)	—	—	—	Experimental – Hollow Fiber – DCMD – Crossflow
Gilron *et al.* (2007)	X	X	X	Simulation – Hollow fiber – DCMD Cascade
Lee *et al.* (2011)	X	X	X	Experimental/Simulation – Hollow Fiber – DCMD Cascade
Hausmann *et al.* (2012)	—	X	X	Experimental/Simulation – Flat Fiber
He *et al.* (2013)	X	X	X	Simulation – Hollow fiber – DCMD Cascade
Chung *et al.* (2014)	X	X	X	Experimental/Simulation – Hollow fiber – DCMD cascade
Duong *et al.* (2015)	—	X	X	Experimental/Simulation – Flat Fiber – DCMD
Lin *et al.* (2014)	X	X	—	Simulation – DCMD – FLR/DLR
Guan *et al.* (2015)	X	X	X	Simulation – Hollow fiber – DCMD
Zhang *et al.* (2015)	X	X	X	Review on heat recovery in MD systems
Lokare *et al.* (2018)	—	X	X	Simulation – DCMD – Brine recycling
Christie *et al.* (2020)	X	X	X	Simulation – DCMD – FLR/DLR
Lisboa *et al.* (2021)	X	—	—	Simulation – Hollow fiber – DCMD
Sampaio (2022)	X	X	—	Simulation – Hollow fiber – DCMD

removed from the system (as a reject stream) or it can be recycled (recirculated). In the present work, the computational code was built taking into consideration both possibilities, simultaneously or not, depending on a flow rejection fraction (λ) at state 9 (c_{s9}, V_9, T_9), which directly and dynamically impacts the concentration of

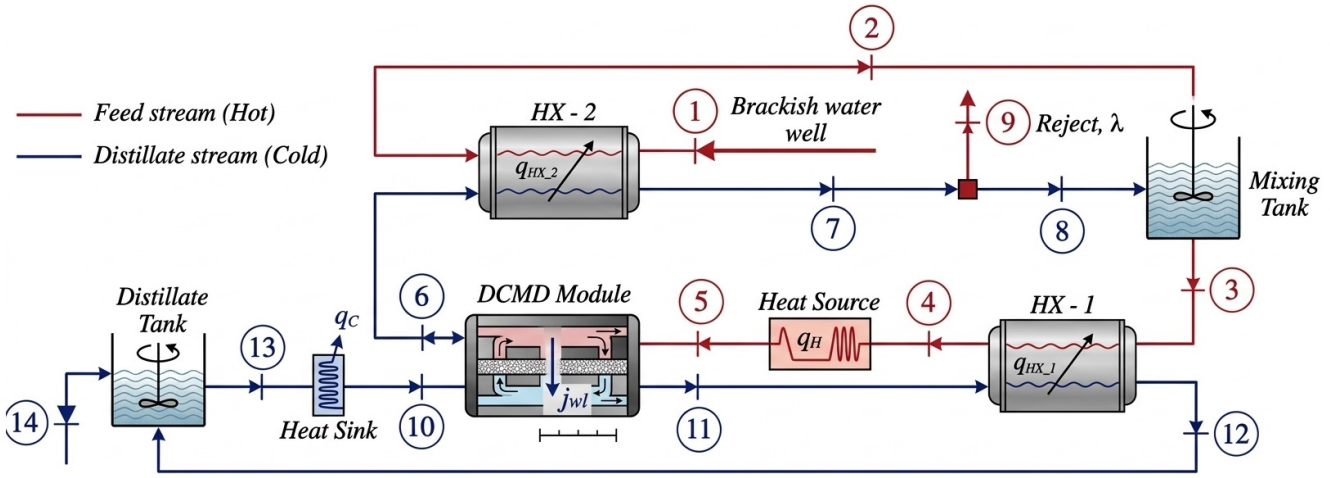


Figure 1: Schematic representation of the DCMD system with double heat recovery and partial or total recirculation of the retentate stream.

the solution in the tank and its volume, V_{mix}. When $\lambda = 1$, the brine is completely rejected at state 9 (the MD system must be continuously fed from the well on a permanent basis). Otherwise, when $\lambda < 1$, the retentate is partially recycled/rejected. Under this condition, unlike common practice, where a steady-state approach is employed for long duration operations, a fully transient operation/simulation must be adopted since the volume and concentration of the mixing tank vary dynamically with time.

When necessary, depending on salinity and volume of the mixing tank (or the operating regime), brackish water can be pumped from the well at state 1 (V_{well}, c_s^{well}, T_{well}), passing through HX-2 as a cold stream to gain energy carried by the effluent retentate from the DCMD module at state 6, which will be cooled down to state 7 (c_{s7}, Q_7, T_7). The stream at state 7 can be directed, either totally ($\lambda = 0$) or partially ($0 < \lambda < 1$), to the mixing tank through state 8 (c_{s8}, V_8, T_8), to be mixed with the heated brackish water stream coming from state 2.

Thus, in addition to the desalination-system operating modes, batch, semi-batch, volume- or salinity-controlled, continuous, and steady-state operation, the numerical simulation will be used to investigate the energy efficiency of the system when an additional heat exchanger, HX-2, is employed. This heat exchanger enables thermal exchange between the retentate stream and the brackish water supplied by the well. The simulation will also evaluate the partial rejection or recirculation of the retentate

stream after heat recovery.

To extend the scope of the simulation, a parametric analysis will be performed to evaluate the interaction between the NaCl concentration in the feed stream and the amount of heat recovered from the fluid returning through the feed circuit of the desalination module. This analysis will also support the development of operational control strategies aimed at maximizing desalination efficiency while reducing brackish-water consumption from the well. To perform the present study, the operational ranges of the PLICA-7 DCMD module produced by Aquastill for experiments in a relevant environment are adopted. Thus, the computational code developed will seek to predict the situations that can happen in experiments with the pilot unit purchased from Aquastill, from small modifications in the installed system, or directly in the laboratory scale experimental module developed in LabMEMS/UFRJ.

3 Mathematical Model

The implemented mathematical model incorporates heat and mass transfer in all system components: mixing tank, distillate tank, DCMD module, heat source and sink, and heat exchangers 1 and 2. For the DCMD module, the model proposed by Lisboa et al. [2021] was extended to the unsteady-state regime. The authors treated the hollow-fiber DCMD module as an equivalent porous medium composed of the following constituents: the porous shell medium (where the feed stream flows), the porous lumen medium (where the distillate stream flows), and the membrane (where the distillation process occurs). The membrane is modeled separately and is not included in the porous medium model. The counterflow configuration for the flow in the hollow fiber is adopted in the present work. An illustrative figure of the porous media model employed to analyze the heat and mass transfer in the vessel is shown in Figure 2.

The hollow-fiber membranes are assumed to be uniformly distributed over the vessel (module) cross-section, with their axes parallel to the vessel axis. The module porosity is defined as the ratio of the phase volume to the total vessel volume:

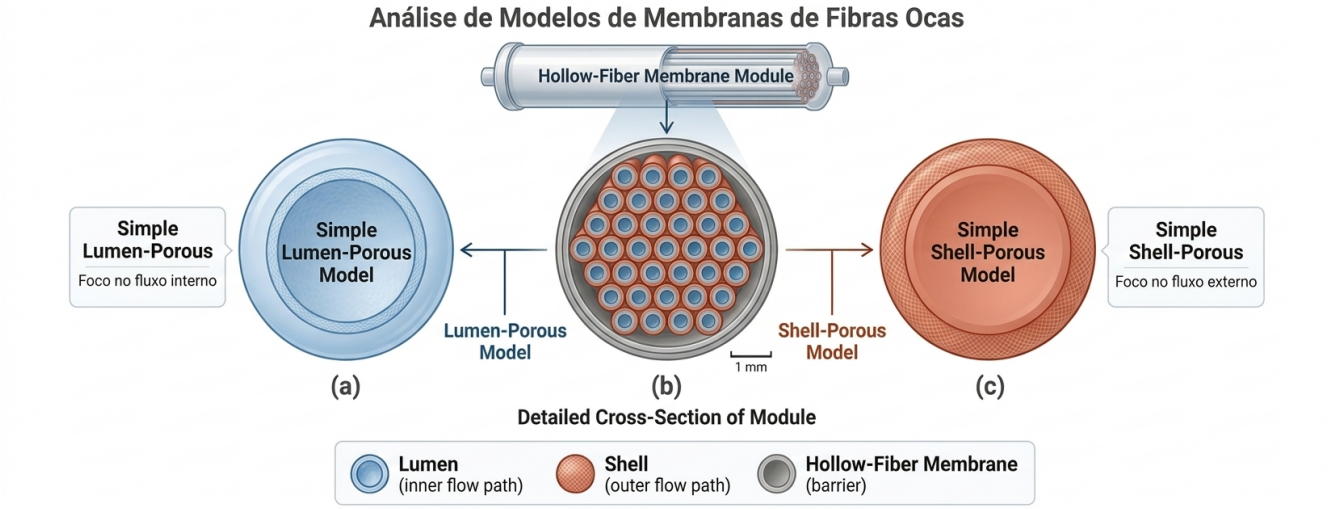


Figure 2: Illustration of the porous media models. Cross sections of the: (a) Porous medium, lumen side; (b) Module and hollow-fiber membrane bundle; (c) Porous medium, shell side.

$$\varepsilon_\gamma = m + (-1)^m N_{\text{fbr}} \frac{r_\gamma^2}{r_v^2}, \tag{1}$$

$$m = \begin{cases} 0, & \text{if } \gamma = l, \\ 1, & \text{if } \gamma = s. \end{cases} \tag{2}$$

Here, ε_γ is the porosity of phase γ (with $\gamma = l$ for the lumen phase and $\gamma = s$ for the shell phase), N_{fbr} is the number of fibers in the vessel, r_γ is the radius of the γ phase, and r_v is the radius of the vessel. Given that the membrane has a thickness δ, the outer radius of the membrane will be $r_s = r_l + \delta$. In the present model, the shell radius, $r_c = r_v/\sqrt{N_{fbr}}$, is an idealization in the simulations for the outer radius of a single membrane of the hollow-fiber module (Figure 3).

The specific surface area, a_γ, used in heat and mass transfer calculations, is the ratio of the surface area available for transport phenomena in each phase to the vessel volume:

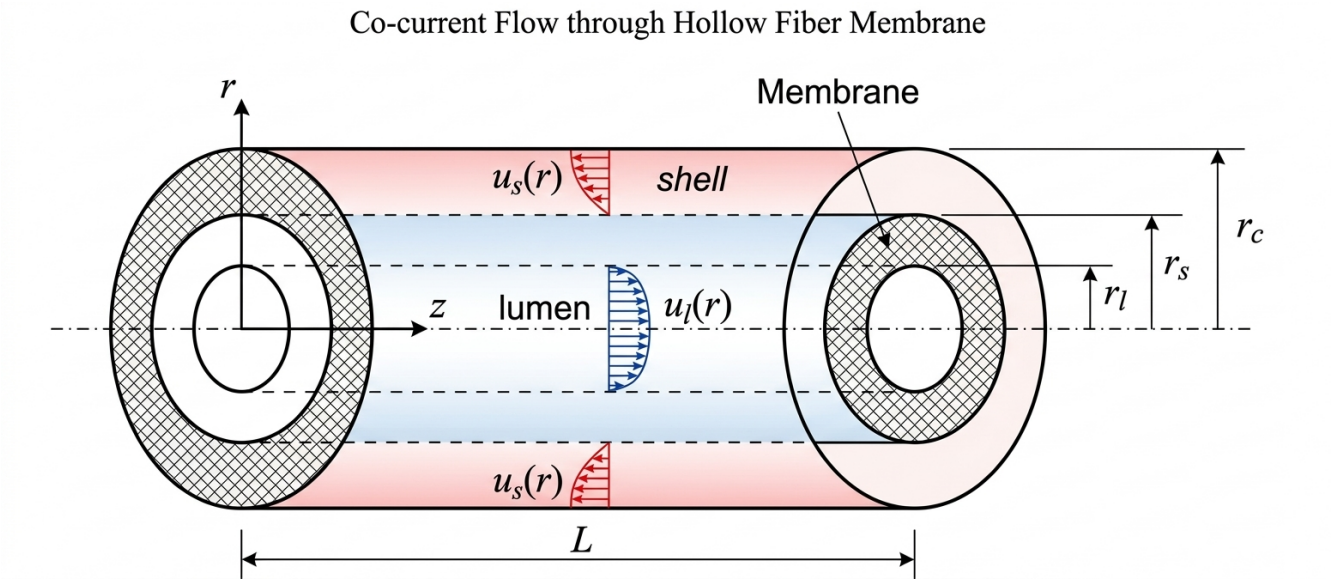


Figure 3: Representation for the single-fiber model of the hollow-fiber membrane module.

$$a_\gamma = \frac{N_{fbr}\, 2\pi r_\gamma}{\pi r_v^2} \tag{3}$$

The physical model for the control volume of the mixing tank considers two inlet streams (retentate stream (8), and brackish well water stream (2)) and the tank outlet stream (3). To recover the heat from the retentate stream, it would be possible to mix it with the well brackish water directly in the mixing tank, and in order to control the salinity in the tank, part of the mixture would be rejected immediately afterwards. This would be the most energetically efficient option, however, some of the brackish water from the well would then be rejected in this process, which is undesirable due to the hydrological stress in arid and semi-arid regions.

Therefore, to avoid this problem, and still recover a large part of the thermal potential of the circulation stream, it is proposed to use an additional heat exchanger (HX-2) before the rejection process (as illustrated in Figure 1), which can be partial or not. Thus, the mixing tank (and therefore all the desalination system) will be a dynamic system, since the volume, V_{mix}, the concentration, $c_{s,mix}$ (or salinity, ϕ_{mix}) and the temperature, T_{mix}, of the mixture inside it change over time.

Considering that the heat exchangers, the distillate and mixing tanks, and the

DCMD module are adiabatic, and that the salinity and temperature fields inside them are uniform (except inside the DCMD module), the physical situation is mathematically modelled as follows.

Mixture and Distillate Tanks

$$\frac{dV_{\text{mix}}}{dt} = \dot{V}_2 + \dot{V}_8 - \dot{V}_3, \qquad t > 0 \tag{4}$$

$$V_{\text{mix}} \frac{df_{\text{mix}}}{dt} = f_2 \dot{V}_2 + f_8 \dot{V}_8 - f_{\text{mix}} \left(\dot{V}_2 + \dot{V}_8 \right), \qquad t > 0 \tag{5}$$

$$\rho_{\text{mix}} V_{\text{mix}} c_{p,\text{mix}} \frac{dT_{\text{mix}}}{dt} = \rho_2 \dot{V}_2 c_{p,2} T_2 + \rho_8 \dot{V}_8 c_{p,8} T_8 - c_{p,\text{mix}} T_{\text{mix}} \left(\rho_2 \dot{V}_2 + \rho_8 \dot{V}_8 \right), \qquad t > 0 \tag{6}$$

$$\frac{dV_{\text{col}}}{dt} = \dot{V}_{12} - \dot{V}_{13} - \dot{V}_{14}, \qquad t > 0 \tag{7}$$

DCMD Module

$$\rho_l c_{p,l} \varepsilon_l \frac{\partial T_l}{\partial t} + \rho_l c_{p,l} u_l \frac{\partial T_l}{\partial z} = -a_l h_l \left(T_l - T_{m,l} \right), \qquad t > 0, \quad \text{lumen side} \tag{8}$$

$$\rho_s c_{p,s} \varepsilon_s \frac{\partial T_s}{\partial t} + \rho_s c_{p,s} u_s \frac{\partial T_s}{\partial z} = -a_s h_s \left(T_s - T_{m,s} \right), \qquad t > 0, \quad \text{shell side} \tag{9}$$

Equations (4)–(9) are subject to the following initial conditions:

$$t = 0: \qquad \begin{cases} V_{\text{mix}}(0) = V_{\text{mix},0}, & f_{\text{mix}}(0) = f_{\text{mix},0}, & T_{\text{mix}}(0) = T_{\text{amb}}, \\ V_{\text{col}}(0) = V_{\text{col},0}, & T_l(z,0) = T_{\text{amb}}, & T_s(z,0) = T_{\text{amb}}. \end{cases} \tag{10}$$

Additionally, since Equations (8) and (9) are partial differential equations, they are subject to the following boundary conditions:

$$T_l(0,t) = T_{e,l} = T_{l,0}, \qquad z = 0, \qquad t > 0 \tag{11}$$

$$T_s(L,t) = T_{e,s} = T_5, \qquad z = L, \qquad t > 0 \tag{12}$$

In these expressions, ρ_γ is the density of phase γ, $c_{p,\gamma}$ is the specific heat at constant pressure, k_γ is the thermal conductivity, T_γ is the bulk temperature, $T_{m,\gamma}$ is the interface temperature between the hollow fiber membrane surface and the liquid phase γ, and h_γ is the convection heat transfer coefficient. The convective heat transfer coefficients for the hot and cold streams are evaluated through the following empirical correlations:

Heat-Transfer Correlations

The Nusselt number for phase γ is calculated as

$$Nu_\gamma = \frac{2r_\gamma h_\gamma}{k_\gamma} = \begin{cases} \left[4.36 + \dfrac{0.036\, Re_\gamma Pr_\gamma \left(d_{h,\gamma}/L\right)}{1 + 0.0011\left[Re_\gamma Pr_\gamma \left(d_{h,\gamma}/L\right)\right]^{2/3}}\right] & \\ \qquad \times \left[\dfrac{1 + N_{\text{fbr}}\left(r_s/r_v\right)}{1 - N_{\text{fbr}}\left(r_s/r_v\right)^2}\dfrac{r_s}{r_v}\right]^m, & Re_\gamma < 2100, \\ \dfrac{\left(Re_\gamma - 1000\right) Pr_\gamma \left(f_\gamma/8\right)}{1.07 + 12.7\left(Pr_\gamma^{2/3} - 1\right)\left(f_\gamma/8\right)^{1/2}} & \\ \qquad \times \left[\dfrac{1 + N_{\text{fbr}}\left(r_s/r_v\right)}{1 - N_{\text{fbr}}\left(r_s/r_v\right)^2}\dfrac{r_s}{r_v}\right]^m, & Re_\gamma \geq 2100. \end{cases} \tag{13}$$

Here, Re_γ is the Reynolds number based on the hydraulic diameter $d_{h,\gamma}$, Pr_γ is the Prandtl number, and f_γ is the friction factor. All these parameters are associated with phase γ and are defined as follows:

$$Re_\gamma = \frac{\rho_\gamma u_\gamma d_{h,\gamma}}{\mu_\gamma} \tag{14}$$

$$Pr_\gamma = \frac{\mu_\gamma c_{p,\gamma}}{k_\gamma} \tag{15}$$

$$d_{h,\gamma} = \begin{cases} 2r_l, & \gamma = l, \\ 2r_v \left[\dfrac{1 - N_{\text{fbr}} \left(r_s / r_v \right)^2}{1 + N_{\text{fbr}} \left(r_s / r_v \right)} \right], & \gamma = s. \end{cases} \tag{16}$$

$$f_\gamma = 2 \left[1.82 \log \left(Re_\gamma \right) - 1.64 \right]^{-2} \tag{17}$$

These equations are also coupled with the additional mass- and energy-balance equations for the remaining control volumes in the feed and distillate streams. Thus, considering that the temperatures T_4 and T_{12}, associated with HX-1, and the temperatures T_2 and T_7, associated with HX-2, are obtained from the definitions of the heat-exchanger effectivenesses, the following relationships are obtained.

Temperatures

$$T_1 = T_{\text{well}} \qquad \text{(input)} \tag{18}$$

$$T_2 = \frac{1}{c_{p2}} \left[c_{p1} T_1 + \varepsilon_{\text{HX},2} \frac{C_{\text{MIN},2}}{\rho_1 \dot{V}_1} \left(T_6 - T_1 \right) \right] \tag{19}$$

$$T_3 = T_{\text{mix}} \tag{20}$$

$$T_4 = \frac{1}{c_{p4}} \left[c_{p3} T_3 + \varepsilon_{\text{HX},1} \frac{C_{\text{MIN},1}}{\rho_3 \dot{V}_3} \left(T_{11} - T_3 \right) \right] \tag{21}$$

$$T_5 = T_{e_s} \qquad \text{(input)} \tag{22}$$

$$T_6 = T_s(0, t) \tag{23}$$

$$T_9 = T_8 = T_7 = \frac{1}{c_{p,7}} \left[c_{p,6} T_6 - \varepsilon_{\text{HX},2} \frac{C_{\text{min},2}}{r_6 \dot{V}_6} \left(T_6 - T_1 \right) \right] \tag{24}$$

$$T_{10} = T_{e,l}, \qquad \text{input} \tag{25}$$

$$T_{11} = T_l(L, t) \tag{26}$$

$$T_{14} = T_{13} = T_{12} = \frac{1}{c_{p,12}} \left[c_{p,11} T_{11} - \varepsilon_{\text{HX},1} \frac{C_{\text{min},1}}{r_{11} \dot{V}_{11}} \left(T_{11} - T_3 \right) \right] \tag{27}$$

Flow Rates

$$\dot{V}_1 = \dot{V}_2 = \dot{V}_{\text{well}}, \qquad \text{input} \tag{28}$$

$$\dot{V}_5 = \dot{V}_4 = \dot{V}_3 = \dot{V}_s, \qquad \text{input} \tag{29}$$

$$\dot{V}_7 = \dot{V}_6 = \left[\left(\varepsilon_m - 1\right) f_5 + 1\right] \dot{V}_5 - \dot{V}_{jwl} \tag{30}$$

$$\dot{V}_8 = \left(1 - l_{\text{rej}}\right) \dot{V}_7 \tag{31}$$

$$\dot{V}_9 = \dot{V}_{\text{rej}} = l_{\text{rej}} \dot{V}_7 \tag{32}$$

$$\dot{V}_{10} = \dot{V}_l, \qquad \text{input} \tag{33}$$

$$\dot{V}_{12} = \dot{V}_{11} = \dot{V}_{10} + \left(1 - \varepsilon_m\right) f_5 \dot{V}_5 + \dot{V}_{jwl} \tag{34}$$

$$\dot{V}_{13} = \left(1 - l_{\text{col}}\right) \dot{V}_{12} \tag{35}$$

$$\dot{V}_{14} = l_{\text{col}} \dot{V}_{12} \tag{36}$$

$$\dot{V}_{jwl} = \frac{j_{wl} A_{\text{memb}}}{r_{\text{H}_2\text{O}}} \tag{37}$$

Salinities

$$f_1 = f_2 = f_{\text{well}}, \qquad \text{input} \tag{38}$$

$$f_5 = f_4 = f_3 = f_{\text{mix}} \tag{39}$$

$$f_9 = f_8 = f_7 = f_6 = a_{\text{memb}} f_5 \frac{\dot{V}_5}{\dot{V}_6} \tag{40}$$

$$f_{10} = 0, \qquad \text{for } a_{\text{memb}} = 1, \qquad \text{input} \tag{41}$$

$$f_{14} = f_{13} = f_{12} = f_{11} = \frac{f_{10}\dot{V}_{10} + (1 - a_{\text{memb}})\, f_5 \dot{V}_5}{\dot{V}_{11}} \tag{42}$$

In the salinity equations, α_{memb} is the membrane effectiveness in retaining salt, thereby preventing salt from passing into the distillate stream. In the present study, it was assumed to be equal to 100%, i.e., $\alpha_{\text{memb}} = 1$. The properties of the brackish well water are determined at each point of the system as functions of the local temperature and salinity [Sparrow, 2003].

The membrane-interface temperatures along the axial direction of the DCMD module, $T_{m,l}(z,t)$ and $T_{m,s}(z,t)$, required in Equations (8) and (9), are evaluated iteratively and dynamically within the numerical routine used to solve the system of differential equations. These temperatures are obtained by equating the local heat-transfer rates in the feed, membrane, and distillate regions.The local convective heat-transfer rate from the shell-side stream to the membrane surface is:

$$\dot{Q}_s = 2\pi r_s \Delta z\, h_s \left(T_s - T_{m,s}\right). \tag{43}$$

The total heat-transfer rate through the membrane is the sum of the conductive and latent heat-transfer rates:

$$\dot{Q}_{\text{memb}} = \dot{Q}_{\text{cond}} + \dot{Q}_{\text{vap}}. \tag{44}$$

The conductive heat-transfer rate through the cylindrical membrane is

$$\dot{Q}_{\text{cond}} = \frac{2\pi \overline{k}_m \Delta z}{\ln\left(r_s/r_l\right)} \left(T_{m,s} - T_{m,l}\right). \tag{45}$$

The latent heat-transfer rate associated with the water-vapor flux is

$$\dot{Q}_{\text{vap}} = 2\pi r_l \Delta z \, j_{wl} h_{fg}. \tag{46}$$

The local convective heat-transfer rate from the membrane surface to the lumen-side stream is

$$\dot{Q}_l = 2\pi r_l \Delta z \, h_l \left(T_{m,l} - T_l\right). \tag{47}$$

By combining Equations (43)–(47), the shell-side membrane interface temperature is obtained as

$$T_{m,s} = T_s - \left[1 + \frac{r_s h_s \left(T_{m,s} - T_{m,l}\right) \ln\left(r_s/r_l\right)}{\overline{k}_m \left(T_{m,s} - T_{m,l}\right) - j_{wl} h_{fg} r_l \ln\left(r_s/r_l\right)}\right]^{-1} \times \left(T_s - T_l\right). \tag{48}$$

Similarly, the lumen-side membrane interface temperature is given by

$$T_{m,l} = T_l + \left[1 + \frac{r_l h_l \left(T_{m,s} - T_{m,l}\right) \ln\left(r_s/r_l\right)}{\overline{k}_m \left(T_{m,s} - T_{m,l}\right) - j_{wl} h_{fg} r_l \ln\left(r_s/r_l\right)}\right]^{-1} \times \left(T_s - T_l\right). \tag{49}$$

The density of the mixture at each position or control volume of the circuit, ρ_i, is calculated from the local salt mass fraction, f_i, using the densities of pure water and salt at $T_{\text{amb}} = 25\,^{\circ}\text{C}$. The reference values employed are $\rho_{\text{salt}} = 2165\ \text{kg}\,\text{m}^{-3}$ and $\rho_{\text{H}_2\text{O}} = 997.4\ \text{kg}\,\text{m}^{-3}$. Thus,

$$\rho_i = (1 - f_i)\,\rho_{\text{H}_2\text{O}} + f_i \rho_{\text{salt}}. \tag{50}$$

Other thermophysical properties, including the specific heat at constant pressure, thermal conductivity, dynamic viscosity, and enthalpy of vaporization, are also evaluated as functions of salinity and temperature. These properties are calculated using the correlations proposed by Sparrow [2003] or by interpolation between values available in thermophysical-property tables. The model used to calculate the effective thermal conductivity of the membrane, $\overline{k}_m$, considering both the polymer matrix and the water vapor contained within the membrane pores, is presented subsequently.

3.1 Modeling Mass Transfer in the Membrane

To partially close the system of equations, the mass-transfer process inside the DCMD module must be modeled to determine the water-vapor mass flux through the membrane, j_{wl}. Following the procedure adopted by Lisboa et al. [2021], the Dusty Gas Model proposed by **?**, one of the most widely used models for predicting distilled-water transport through porous membranes, is employed. The molar transport of water vapor through the membrane pores is described by

$$\frac{N_w}{D_w^k} + \frac{p_a N_w - p_w N_a}{D_{wa}^0} = -\frac{1}{RT_m}\nabla p_w, \tag{51}$$

where N_w is the molar flux of water vapor, N_a is the molar flux of air, p_a is the partial pressure of air inside the membrane pores, p_w is the partial pressure of water vapor, D_w^k is the Knudsen diffusivity of water vapor, D_{wa}^0 is the molecular diffusivity of water vapor in air, R is the universal gas constant, and T_m is the average thermodynamic temperature at each axial position along the membrane.

Air is assumed to remain stationary inside the membrane pores, such that

$$N_a \approx 0. \tag{52}$$

Applying Dalton's law, the molar flux of water vapor through the hollow-fiber membrane, based on the inner membrane radius, is given by

$$N_{wl} = \frac{D^0_{wa}}{RT_m r_l \ln\left(r_s/r_l\right)} \ln\left[\frac{D^0_{wa} - D_{\text{eff}} p_s}{D^0_{wa} - D_{\text{eff}} p_l}\right]. \tag{53}$$

The corresponding water-vapor mass flux is

$$j_{wl} = M_w N_{wl}, \tag{54}$$

where M_w is the molar mass of water.

The molecular diffusivity of water vapor in air is calculated from

$$D^0_{wa} = 4.46 \times 10^{-6} \frac{\varepsilon_m}{\tau_m} T_m^{2.334}. \tag{55}$$

The Knudsen diffusivity of water vapor is given by

$$D^k_w = \frac{\varepsilon_m}{\tau_m} \frac{d_p}{3} \sqrt{\frac{8RT_m}{\pi M_w}}. \tag{56}$$

The effective diffusivity resulting from the combined molecular and Knudsen transport mechanisms is

$$D_{\text{eff}} = \frac{D^k_w D^0_{wa}}{D^0_{wa} + P D^k_w}. \tag{57}$$

The total pressure inside the membrane pores, P, is assumed to be equal to atmospheric pressure. The water-vapor partial pressures at the shell-side and lumen-side

membrane interfaces, p_s and p_l, respectively, are calculated from the corresponding saturation vapor pressures. The effect of salinity is included through the salt mole fraction, $x_{\text{salt},\gamma}$, and the water activity coefficient, a_γ, according to

$$p_\gamma = (1 - x_{\text{salt},\gamma})\, a_\gamma p_{v,\gamma}, \qquad \gamma \in \{s, l\}, \tag{58}$$

where $p_{v,\gamma}$ is the saturation vapor pressure evaluated at the corresponding membrane-interface temperature using the Antoine equation. The activity coefficient is calculated as

$$a_\gamma = 1 - 0.5 x_{\text{salt},\gamma} - 10 x_{\text{salt},\gamma}^2. \tag{59}$$

The morphological characteristics of the membrane are included in the calculation of the distillate mass flux through the mean pore diameter, d_p, porosity, ε_m, and tortuosity, τ_m.

Models for the effective membrane thermal conductivity, $\overline{k}_m$, and the membrane tortuosity, τ_m, must also be selected to close the governing equations. The most common effective-thermal-conductivity models are based on different geometrical arrangements of the solid and gaseous phases. In the parallel model, the solid and gaseous phases are arranged in parallel:

$$\overline{k}_m = (1 - \varepsilon_m)\, \overline{k}_s + \varepsilon_m \overline{k}_f, \qquad \text{parallel model.} \tag{60}$$

In the series model, the two phases are arranged in series:

$$\overline{k}_m = \left[\frac{1 - \varepsilon_m}{\overline{k}_s} + \frac{\varepsilon_m}{\overline{k}_f} \right]^{-1}, \qquad \text{series model.} \tag{61}$$

For porous media with porosity greater than approximately 0.6, **?** recommend the Maxwell I model. This model assumes that the polymeric matrix consists of a random distribution of solid particles that do not interact with one another within

a homogeneous fluid medium. The resulting effective conductivity is

$$\overline{k}_m = \overline{k}_f \left[\frac{1 + 2\left(1 - \varepsilon_m\right) b_{sf}}{1 - \left(1 - \varepsilon_m\right) b_{sf}} \right], \qquad \text{Maxwell I model,} \tag{62}$$

where

$$b_{sf} = \frac{\overline{k}_s - \overline{k}_f}{\overline{k}_s + 2\overline{k}_f}. \tag{63}$$

The manner in which the pores and solid matrix are distributed within the membrane governs the diffusion path and, consequently, the vapor flux. Membrane tortuosity, defined as the ratio between the actual diffusion path length and the membrane thickness, is therefore an important parameter for predicting the transmembrane mass flux.

Several tortuosity models are available in the literature [Ghanbarian et al., 2013]. Kim et al. [2021] analyzed eleven tortuosity models, including eight geometrical models based on Euclidean structures, such as cubes, spheres, and cylinders, and three models based on the fractal theory of porous media. For the geometrical model proposed by **?**, the membrane tortuosity is

$$\tau_m = \frac{\left(2 - \varepsilon_m\right)^2}{\varepsilon_m}. \tag{64}$$

The model proposed by Beeckman [1990] is

$$\tau_m = \frac{\varepsilon_m}{1 - \left(1 - \varepsilon_m\right)^{1/3}}. \tag{65}$$

For the fractal model proposed by Elias-Kohav et al. [1991], the tortuosity is

$$\tau_m = \frac{1}{\varepsilon_m}. \tag{66}$$

The Bruggeman model is given by

$$\tau_m = \frac{1}{\sqrt{\varepsilon_m}}. \tag{67}$$

According to [Curcino et al., 2022], who evaluated several effective-thermal- conductivity and tortuosity models for membrane-distillation simulations in flat-sheet and hollow-fiber configurations, an appropriate model should be selected based on the known morphological and structural properties of the membrane employed in the experimental desalination module.

3.2 Efficiency Metrics

To evaluate the desalination system and quantify improvements in the energy efficiency of the membrane-distillation process, the thermal efficiency, η_{th}, gain-output ratio, GOR, specific thermal energy consumption, SEC_{th}, and water recovery ratio, RR, are calculated under the different operating scenarios. The thermal efficiency of the membrane is defined as:

$$\eta_{th} = \frac{\dot{Q}_{\mathrm{vap}}}{\dot{Q}_{\mathrm{memb}}}. \tag{68}$$

The gain-output ratio is defined as

$$\mathrm{GOR} = \frac{\dot{Q}_{\mathrm{vap}}}{\dot{Q}_{\mathrm{ext}}}. \tag{69}$$

The specific thermal energy consumption is

$$\mathrm{SEC}_{th} = \frac{\dot{Q}_{\mathrm{ext}}}{\dot{V}_{jwl}}. \tag{70}$$

The water recovery ratio is

$$\mathrm{RR} = \frac{\dot{m}_{\mathrm{dist}}}{\dot{m}_{\mathrm{well}}}. \tag{71}$$

The total heat-transfer rate through the membrane is the sum of the conductive and latent heat-transfer rates:

$$\dot{Q}_{\mathrm{memb}} = \dot{Q}_{\mathrm{cond}} + \dot{Q}_{\mathrm{vap}}. \tag{72}$$

The total conductive heat-transfer rate through all hollow fibers is calculated as

$$\dot{Q}_{\mathrm{cond}} = \left| N_{\mathrm{fbr}} \int_0^L \frac{2\pi \overline{k}_m}{\ln\left(r_s/r_l\right)} \left(T_{m,s} - T_{m,l}\right) \, \mathrm{d}z \right|. \tag{73}$$

The total latent heat-transfer rate associated with water-vapor transport is

$$\dot{Q}_{\mathrm{vap}} = \left| N_{\mathrm{fbr}} \int_0^L 2\pi r_l j_{wl} h_{fg} \, \mathrm{d}z \right|. \tag{74}$$

The external heat demand is calculated by subtracting the heat recovered in HX-1 and HX-2 from the total thermal demand:

$$\dot{Q}_{\mathrm{ext}} = \dot{Q}_{\mathrm{tot}} - \left(\dot{Q}_{\mathrm{HX},1} + \dot{Q}_{\mathrm{HX},2} \right). \tag{75}$$

The total thermal demand required to heat the feed stream is

$$\dot{Q}_{\mathrm{tot}} = \rho_3 \dot{V}_3 c_{p,3} \left(T_{e,s} - T_{e,l}\right). \tag{76}$$

The heat-transfer rate recovered in HX-1 is

$$\dot{Q}_{\mathrm{HX},1} = \rho_3 \dot{V}_3 \left(c_{p,4} T_4 - c_{p,3} T_3\right)$$

$$= \rho_{11}\dot{V}_{11}\left(c_{p,11}T_{11} - c_{p,12}T_{12}\right). \quad (77)$$

The heat-transfer rate recovered in HX-2 is

$$\dot{Q}_{\mathrm{HX},2} = \rho_1\dot{V}_1\left(c_{p,2}T_2 - c_{p,1}T_1\right)$$
$$= \rho_6\dot{V}_6\left(c_{p,6}T_6 - c_{p,7}T_7\right). \quad (78)$$

The global GOR and SEC_{th} values are calculated as average values over the complete simulation period. The volumes of distilled water produced and brackish well water consumed are obtained by integrating the corresponding instantaneous volumetric flow rates over time.

A computational code for solving the coupled system of ordinary and partial differential equations governing the temperatures, flow rates, and salinities was developed in Fortran 95. The MMOCLH routine [IMSL Fortran Math Library, 2022], which solves hyperbolic partial differential equations using the method of lines, was employed with an error tolerance of 10^{-5}.

The nonlinear equations used to calculate the membrane-interface temperatures, $T_{m,s}$ and $T_{m,l}$, were solved with a prescribed error tolerance of 10^{-11} using the ZBREN routine [IMSL Fortran Math Library, 2022].

4 The System Operating Modes

To control the operation of the dynamic system and optimize the desalination system water production and brackish water consumption, a general algorithm based on volume and salinity of the mixture in the mixing reservoir is applied at each instant of time. The developed computational code is very flexible, allowing simulations of different desalination system operation scenarios: steady-state regime (no recirculation), batch recirculation, semi-batch recirculation, continuous recirculation (dynamic rejection fraction/recirculation), and partial recirculation operation mode with volume or concentration control.

For the unsteady-state regimes, the system can start the operation with no brine

rejection (total recirculation), $\lambda = \lambda_0 = 0$, or with a predefined rejection (partial recirculation), $\lambda_0 \neq 0$. For these regimes, the desalination system operates in such a way that the mixing tank, initially with a volume of brackish water of $V_{mix,0} = 100$ L, has its mixture consumed by the DCMD module, as distillate is produced over time.

Additionally, to analyze the influence of the external heat exchangers on the system performance it is sufficient to vary their effectiveness (ε_{HX1} and ε_{HX2}). By choosing $\varepsilon_{HX1} = 0$ and/or $\varepsilon_{HX2} = 0$, the heat exchangers are not included in the analysis.

4.1 Batch and semi-batch operations

In batch and semi-batch operation modes, over each process cycle, the distillate is continuously removed from the system, and the remaining solution that leaves the MD module, with higher salinity, is directed to the feed tank ($\lambda = 0$). In the batch operation process, there is no water pumping from the well ($\dot{V}_{well} = 0$) between recycling, so the volume of the solution in the tank decreases, and the salt concentration increases over time. In the semi-batch operation cycle, the volume of the mixing tank is kept constant, by continuously adding fresh feed water from the well to the system, at the same rate at which distillate is removed ($\dot{V}_{well} = \dot{V}_{jwl}$).

In both operation processes, the permeate production rate and the heat-transfer rate decrease as the feed stream of the MD module becomes increasingly saturated with salts. After the salinity of the solution in the mixing tank reaches a limiting value, the high-salinity brine is flushed-out (λ is set to 1) until the minimum allowable volume is reached. Then, the well pump refills the tank from the well ($\dot{V}_{well} = \dot{V}_s$) until the minimum concentration is reached, and the cycle restarts. During these processes (flushing and filling), the HX-2 should be used to improve the system's efficiency.

4.2 Continuous operation (steady-state operation with partial brine recirculation)

In the continuous operation regime, a fixed operational brine salinity is specified at the MD module exit. Therefore, after the initial transients have elapsed and the solution that leaves the module is at the required salinity (reaching the steady-state regime), it must be continuously withdrawn from the system at a specific fraction rejection (λ) that depends on the MD module performance. The make-up water from the well ($\dot{V}_{well}$) is simultaneously added to the tank at a proportional flow rate that maintains the salinity of the stream at the MD module exit, which is associated with the corresponding rejection fraction. Again, the HX-2 should be used to improve the system's efficiency. The rejection fraction and the volumetric flow rate from the well are obtained from the following salt and total-mass balances applied to the mixing tank:

$$\lambda = \frac{(f_3 - f_2)\,\dot{V}_{\text{well}} + (f_3 - f_8)\,\dot{V}_7}{(f_2 - f_8)\,\dot{V}_7} \tag{79}$$

$$\dot{V}_{well} = \dot{V}_{jwl} + \lambda \left(\dot{V}_s - \dot{V}_{jwl} \right) \tag{80}$$

4.3 Steady-state operation with no brine recirculation

To simulate a steady-state condition with no recirculation, it is sufficient to set $\lambda = \lambda_0 = \lambda_1 = 1$ and $\dot{V}_{well} = \dot{V}_s$ all the time during the simulations. In this situation, all retentate leaving the DCMD module is rejected and the system is continuously fed by the well at constant feed flow rate and salinity. As will be shown, if this mode of operation is mandatory, as in the case studied by Sampaio [2022], the use of the second heat exchanger (HX-2) to recover heat from the retentate stream before it is flushed out leads to an improvement in the performance of the desalination system.

4.4 Volume or Concentration controlling

Because these operating modes are inherently transient, the distillate is continuously withdrawn from the system, while the concentrated stream leaving the MD module, whose salinity progressively increases, is either recirculated to the mixing tank or discharged from the system, depending on the value of λ. In both cases, the use of HX-2 to recover thermal energy from the retentate stream before its discharge improves the overall energy efficiency of the desalination system, as demonstrated later.

In the volume-controlled mixing-tank operating mode, predefined minimum and maximum tank volumes are established as operational limits. As distillate is produced, the salinity of the solution in the mixing tank, $c_{s,\mathrm{mix}}(t)$, continuously increases, whereas the liquid volume, $V_{\mathrm{mix}}(t)$, decreases. When the minimum allowable volume, V_{min}, is reached, brackish water from the well is supplied to the tank until the maximum predefined volume, V_{max}, is restored. This cycle is repeated throughout the desalination process. If the solution in the mixing tank reaches a predefined maximum operating salinity, the tank contents are completely discharged, corresponding to $\lambda = 1$. The tank is then refilled with brackish well water until the initial volume, $V_{\mathrm{mix},0}$, is restored.

On the other hand, in the mixing tank salinity controlling operation, a minimum and a maximum allowed mixing tank salinity is set as the operation limits. The salinity of the water in the mixing tank ($c_{s,\mathrm{mix}}(t)$) is continuously increasing with the distillate production, while its volume ($V_{\mathrm{mix}}(t)$) decreases since there is no brackish water flow from the well. At the instant the maximum salinity allowed is reached ($C_{s,\mathrm{max}}$), the pump from the well is turned on to feed the mixing tank until the salinity of the water contained in it falls to the minimum predefined value. In this process, the volume of the mixing tank is increased, so its operational range will be dynamically adjusted according to the values of salinity limits previously set.

5 Results

5.1 Algorithm verification

Before analyzing typical well operation scenarios, the developed computational code is verified by comparing its results with available literature data. Since no transient results are available, these comparisons are made with data for the steady-state regime. Table 2 presents the physical and geometric characteristics of the membranes and modules employed in the comparisons. The experimental results of Wang et al. [2008], Yang et al. [2011] and numerical results of Guan et al. [2014] and Kim [2014] are employed for situations with no heat exchangers, i.e., only the performance of the DCMD module is studied.

It should be noted that the experiments reported by Wang et al. [2008] were conducted under a co-current flow arrangement, whereas the present study adopts a counter-current configuration. Nevertheless, the numerical investigation performed by Kim [2014], which compared both flow arrangements, indicated that the differences in the predicted performance were small within the range of operating parameters considered.

It is also important to note that, although the simulations presented by Guan et al. [2014] were performed under a counter-current flow arrangement, they adopted the HICO configuration, with the hot fluid flowing through the lumen and the cold fluid through the shell. In contrast, the present work employs the HOCI configuration, in which the hot fluid flows through the shell and the cold fluid through the lumen. According to Guan et al. [2012, 2014], based on the conclusions of Wirth and Cabassud [2002], no significant differences are expected between the HICO and HOCI configurations provided that similar hydrodynamic conditions are maintained in both fluid streams.

Table 3 presents a verification of the results obtained with the present work together with the work of Lisboa et al. [2021] and a comparison with the experimental results of Wang et al. [2008], for the outlet temperatures of the feed and distillate streams, as well as the distillate flow. The results show that the present methodology is consistent with the analyzed physics, at least in the steady-state regime, when

Table 2: Characteristics (input data) of the membranes and modules used for comparison.

Parameter	Symbol	Unit	Wang *et al.* (2008)	Kim (2014)	Yang *et al.* (2011)	Guan *et al.* (2014)
			Experimental	Numerical	Experimental	Numerical
Membrane inner radius	r_l	mm	0.3	0.3	0.46	0.49
Membrane thickness	t	mm	0.11	0.11	0.275	0.240
Polymer thermal conductivity	k_p	W/(m·K)	0.18[a]	0.18	0.18[a]	0.18[a]
Average pore diameter	d_p	mm	0.17[b]	0.17	0.145/0.191	0.2[b]
Porosity	ε_m	—	0.738	0.738	0.83	0.85[b]
Shell inlet temperature	$T_{e,s}$	°C	30–79.3	30–79.3	35–70	30–80
Lumen inlet temperature	$T_{e,l}$	°C	17.5	17.5	25	23.5–30
Total pore pressure	P	kPa	101.325[a]	101.325[a]	101.325[a]	101.325[a]
Shell side flow rate	$\dot{V}_s$	L/min	1.648[c]	1.521	2.5–3.0	1.14–3.0
Lumen side flow rate	$\dot{V}_l$	L/min	0.407[c]	0.407	0.4	0.046–0.4
Number of fibers	N_{fbr}	—	30[a]	30	20	22
Vessel radius	r_v	mm	3.241[a]	3.232	4.75	7.5
Vessel length	L	m	0.200	0.200	0.225	0.200
Salt mass fraction shell	$w_{NaCl,s}$	%	3.5[d]	3.5	3.5	7.0
Lumen salt mass fraction	$w_{NaCl,l}$	%	0.0	0.0	0.0	0.0
Cold/hot fluid arrangement	—	—	HOCI	HOCI	HOCI	HICO
Direction of flow	—	—	Co-current	Counter	Counter	Counter
Membrane area	A_{memb}	cm^2	150	152.7	220	180.9
Packing density	φ	%	48[a]	48	50	20.8[a]

[a] Obtained from, or employed the calculation procedure in, Kim [2014]. [b] Assumed in comparison with other works. [c] Calculated as a function of average velocities ($u_s = 1.6$ m/s and $u_l = 0.8$ m/s) and the respective transversal section areas. [d] Lisboa et al. [2021] used $w_{NaCl,s} = 2.5\%$ in their simulations.

Table 3: Verification of the computational code with the numerical results of Lisboa et al. [2021] and comparison with the experimental data of Wang et al. [2008]. $T_{e,l} = 17.5$°C.

$T_{e,s}$ [°C]	ΔT [°C]	$T_{o,s}$ [°C]		$T_{o,l}$ [°C]		j_{wl} [kg/(m^2·h)]	
		Present	Wang (2008)	Present	Wang (2008)	Present	Wang (2008)
40.0	22.5	38.5	39.0	23.4	24.5	5.5	6.2 ± 0.4
49.5	23.0	47.2	48.0	26.5	27.8	10.1	11.5 ± 0.7
59.8	42.3	56.2	57.4	31.7	31.3	20.0	19.9 ± 1.0
70.2	52.7	65.1	67.2	37.5	36.2	32.8	30.8 ± 1.0
79.3	61.8	72.6	76.0	43.8	39.0	47.9	41.5 ± 1.2

* Based on the outer radius of the hollow fiber. $T_{Wang_et_al.}(2008)$: $T \pm 0.1$°C.

compared to the experimental results of Wang et al. [2008]. This table also shows that the results obtained with the methodology in transient regime converge, as would be expected for long times, to the numerical results of Lisboa et al. [2021], who also verified their code with the experiments of Wang et al. [2008].

Another verification is illustrated in Figure 4, which shows a comparison of the results obtained in the present work for the produced distillate flux (which match the numerical results obtained by Lisboa et al. [2021]) with the experimental results of Wang et al. [2008] and the numerical results of Kim [2014].

The comparison with the results of Wang et al. [2008] is made by employing the parallel model of effective thermal conductivity (Eq. 60) and the fractal model of Elias-Kohav et al. [1991] for tortuosity (Eq. 66), while the comparison with the results of Kim [2014] is made with the Maxwell I model (Eq. 62) of effective thermal conductivity and the model of Beeckman [1990] for tortuosity (Eq. 65). The distillate mass flux was calculated with the surface area based on the outer radius of the hollow fiber.

To illustrate the spatial temperature distribution along the DCMD module, Figure 5 brings a comparison between the results obtained with the present formulation and the numerical results of Guan et al. [2014] for the steady-state temperatures along the feed channel, T_s, of the feed/membrane interface, $T_{m,s}$, of the membrane/distillate interface, $T_{m,l}$, and of the distillate channel, T_l. The inlet temperatures and flow rates of the feed and distillate streams employed in the simulation were: $T_{e,s} = 80°\text{C}$, $T_{e,l} = 30°\text{C}$ and $\dot{V}_s = 1.1425$ L/min, $\dot{V}_l = 0.04635$ L/min, respectively.

Although the effective thermal conductivity and tortuosity models employed by Guan et al. [2014] were not defined in their work, the agreement between the numerical results is satisfactory. Here, the results were obtained by making use of the parallel model for the effective thermal conductivity and the fractal model of Elias-Kohav et al. [1991] for the tortuosity. In addition to these differences, the HICO flow configuration used in the work of those authors differs from the HOCI configuration simulated in the present work, so the small departures from their results were expected.

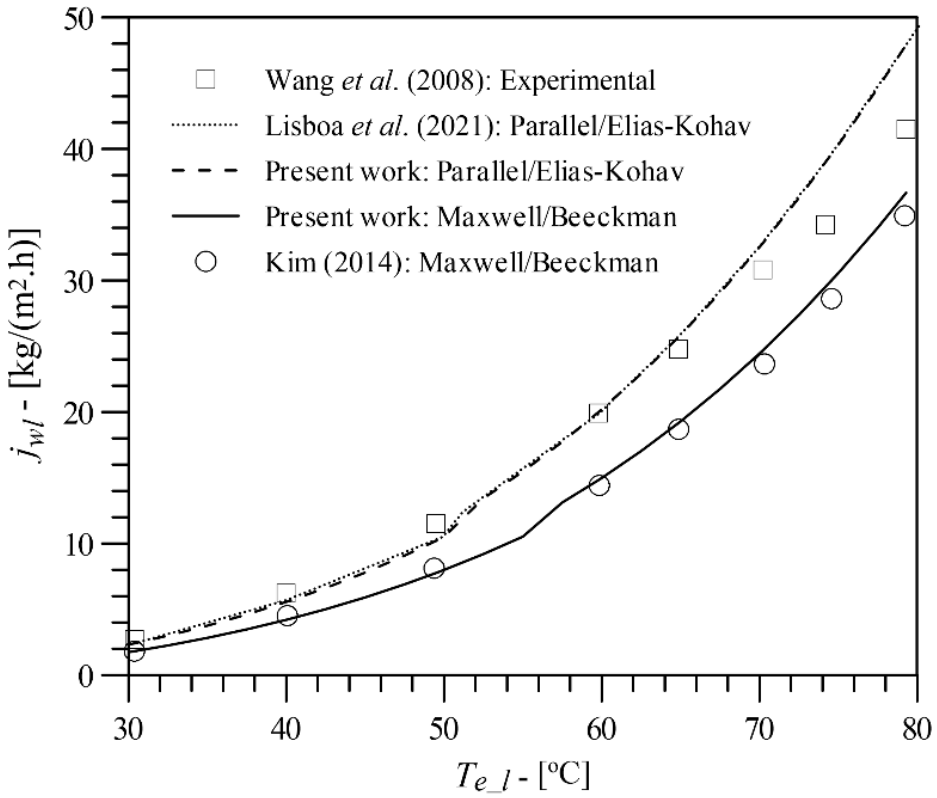


Figure 4: Comparison between the result obtained with the present methodology and the experimental results of Wang et al. [2008] and numerical results of Kim [2014] for the distillate flux.

5.2 Simulations with the Brackish Water Well

For simulation of a real desalination system fed by brackish water wells, data from wells commonly found in the northeastern region of Brazil were used, whose temperatures, salt concentrations, typical flow rates, and other properties can be found in Veiga [2017]. Table 4 provides the characteristic parameter values, which serve as input data for the code, of 444 wells located in the Ceará state of Brazil.

The parameters relating to the characteristics of the DCMD module and the membrane are those provided in Table 5. The flow arrangement of the module is carried out in the HOCI configuration (hot fluid in the shell and cold fluid in the lumen) in counter-current.

The data of the module was obtained from the work of Lisboa et al. [2021], who performed steady-state analyses of a DCMD module with membrane surface area similar to that manufactured by the company Aquastill (PLICA 7 model, membrane area of 7.2 m^2), and installed in the external area of the LabMEMS/UFRJ. Ad-

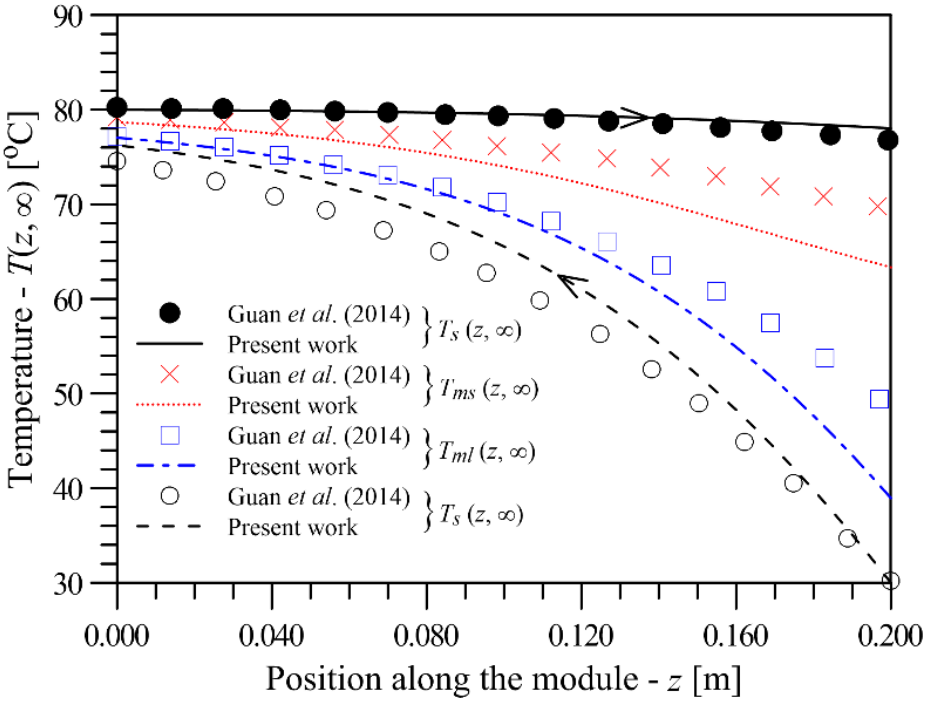


Figure 5: Temperature distribution along the axial position in the DCMD module in the feed, interface, and distillate regions. Comparison with the numerical results of Guan et al. [2014].

Table 4: Characteristics of wells in the state of Ceará/Brazil.

Limits	Salt concentration [g_s/L]	Flow rates	
		[m^3/h]	[L/min]
Maximum	13	15	250
Minimum	0.199	0.10	1.67
Average	2.532	3.45	57.5
Adopted	3.5	3	50

$T_{well} = T_{amb} = 25°C$

ditional parameters were evaluated using the procedure illustrated in Kim [2014] and, based on literature data, adopting values of characteristic membrane parameters (porosity, thickness, among others). All the results presented were obtained by making use of the parallel model for the effective thermal conductivity and the fractal model of Elias-Kohav et al. [1991] for the tortuosity.

Simulations were performed for various scenarios, all of them starting from the condition that the well feeds the system through the mixing tank (initially $V_{mix,0} =$

Table 5: Input data for the membrane and module used in the well simulations.

Parameter	Symbol	Unit	Lisboa *et al.* (2021)
Membrane inner radius	r_l	mm	0.3
Membrane thickness	t	mm	0.1
Polymer thermal conductivity	k_p	W/(m·K)	0.2
Average pore diameter	d_p	mm	0.2
Membrane porosity	ε_m	—	0.8
Shell-side inlet temperature	$T_{e,s}$	°C	80
Lumen-side inlet temperature	$T_{e,l}$	°C	25
Total pore pressure	P	kPa	101.325
Shell-side flow rate	$\dot{V}_s$	L/min	30–50–80
Lumen-side flow rate	$\dot{V}_l$	L/min	30–50–80
Number of fibers	N_{fbr}	—	10,000
Vessel radius	r_v	mm	50
Vessel length	L	m	1
Shell-side salt mass fraction	$w_{NaCl,s}$	%	0.035
Lumen-side salt mass fraction	$w_{NaCl,l}$	%	0
Membrane area	A_{memb}	m^2	7.2
Packing density	φ	%	0.64

100 L and $T_{mix} = 25\,°C$) only when some of the points of the controlling operating mode adopted are reached.

To illustrate a typical dynamic behavior of the heat transfer process inside the DCMD module, Figure 6 shows the bulk temperature profiles of the retentate and distillate streams along the respective channels inside the module, at four instants of time (approximately 2.8×10^{-4} s, 3.4 s, 16 s, and 30 s), for the flow condition $\dot{V}_l = \dot{V}_s = 50$ L/min. The simulation was performed with no recirculation of the brine ($\lambda = 1$), and use of heat exchangers is irrelevant in this type of analysis.

As can be seen from this figure, the time evolution of the heat transfer inside the DCMD module is very fast, since the steady-state condition is reached in just a few seconds, taking no more than 30 s for the module to attain this regime. Therefore, any numerical (experimental) analysis employing large time steps (measurement intervals) will not adequately capture the transient phenomena happening inside the DCMD module.

Figures 7a and 7b illustrate bulk temperature profiles of the feed and permeate

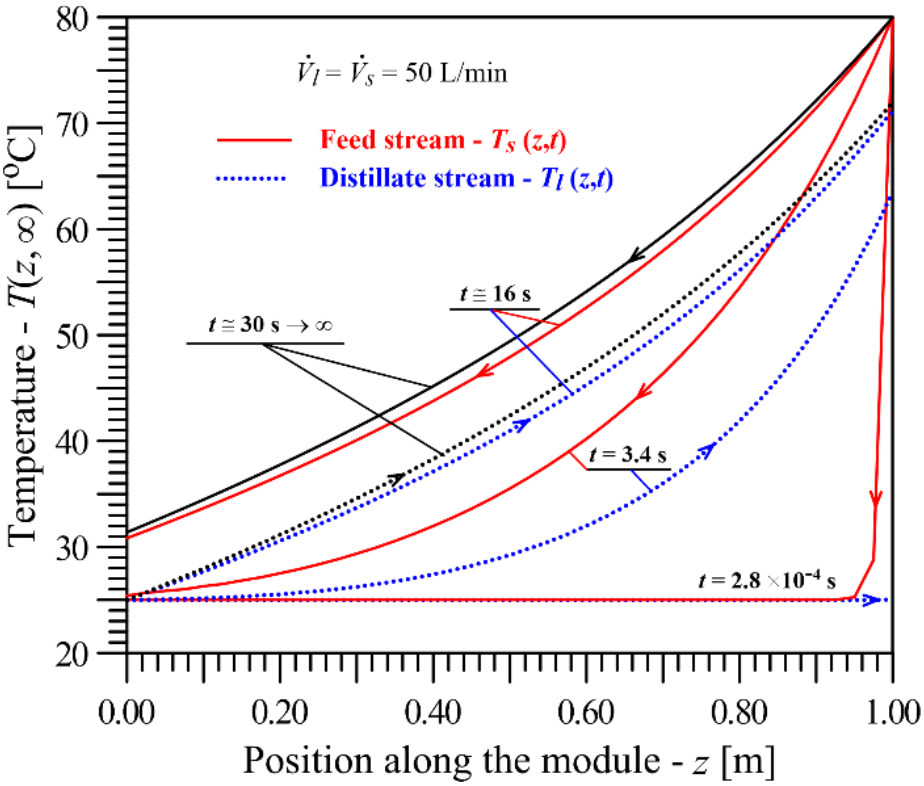


Figure 6: Time evolution of temperature profiles inside the DCMD module for $\dot{V}_l = \dot{V}_s = 50$ L/min.

channels along the module for different conditions of volumetric flow rate of the streams, considering steady-state single-pass flow in the DCMD module, and no heat recovery ($\lambda_0 = \lambda = 1$, $\dot{V}_{well} = \dot{V}_s$, $\varepsilon_{HX1} = \varepsilon_{HX2} = 0$). The objective is to obtain indications of possible flow configurations (well and distillate flow rates) with adequate potential for using heat exchangers to enhance the energy efficiency of the desalination system, as well as to optimize the water recovery (maximize the distillate flux).

Figure 7a illustrates two scenarios where the distillate flow rate is lower than the feed flow rate, and two scenarios where the distillate flow rate is greater than the feed flow rate, approximating what Christie et al. [2020] called the DLR (Distillate Limiting Regime) and the FLR (Feed Limiting Regime) configurations, respectively. Figure 7b shows three cases where the distillate and feed flow rates are equal.

At a first glance, observing the streams outlet temperatures of the DLR configuration on Figure 7a, one could point out that this configuration would have a great potential for energy recovery, since both streams present the highest outlet

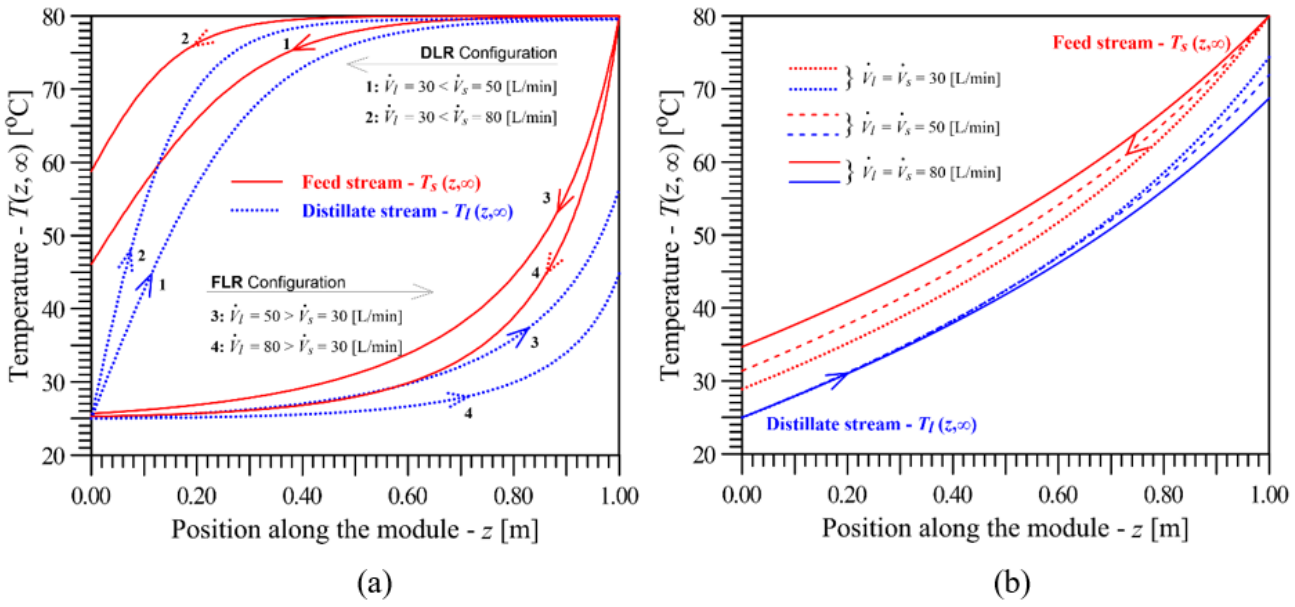


Figure 7: Steady-state temperature profiles along the axial position in the module DCMD for: a) different feed and distillate flow rates; b) equal distillate and feed flow rates ($\dot{V}_l = \dot{V}_s$).

temperatures (79.6 °C for the distillate stream, and 58.8 °C for the feed stream), while in the FLR configuration only the distillate stream presents potential for heat recovery (44.8 °C). However, as it will be shown, the DLR configuration presents the lowest water recovery factor and the highest specific thermal energy cost, becoming inappropriate for heat regeneration.

For an intuitive understanding of how the operational and configurational decisions affect energy efficiency, Christie et al. [2020], based on Lin et al. [2014], introduced the critical mass-flow-rate ratio, α^*, to define the theoretical condition for the maximum water-recovery factor in steady-state, single-pass DCMD systems, $R_{\max}$:

$$R_{\max}^{\mathrm{DLR}} = \frac{\dot{J}_{wl,\mathrm{sys,max}}}{\dot{m}_s} = \frac{\eta_{\mathrm{th}}\, c_{pl}}{\overline{h}_{\mathrm{vap}}}\, \alpha\, \Delta T \tag{81}$$

$$R_{\max}^{\mathrm{FLR}} = \frac{\dot{J}_{wl,\mathrm{sys,max}}}{\dot{m}_s} = \frac{\eta_{\mathrm{th}}\, c_{ps}}{\overline{h}_{\mathrm{vap}}}\, \Delta T \tag{82}$$

$$\alpha = \frac{\dot{m}_l}{\dot{m}_s} \tag{83}$$

$$\alpha^* = \frac{c_{ps}}{c_{pl}} \left(\frac{\overline{h}_{\text{vap}} - \dfrac{c_{ps}\Delta T}{2}}{\overline{h}_{\text{vap}} + \dfrac{c_{pl}\Delta T}{2}} \right) \tag{84}$$

For $\alpha < \alpha^*$, the module operates in the distillate-limited regime (DLR). In this regime, the distillate flow rate is lower than the feed flow rate, and the distillate temperature increases along the module toward the feed-side threshold temperature, $T_{\text{thres},s}$. This threshold is defined as the temperature at which the distillate stream, with $C_s = 0$, has the same partial vapor pressure as the feed stream at the inlet temperature $T_{e,s}$ and salinity C_s. Therefore,

$$p_v\left(C_s, T_{e,s}\right) = p_v\left(0, T_{\text{thres},s}\right), \tag{85}$$

at which point the transmembrane vapor-pressure difference vanishes.

Conversely, for $\alpha > \alpha^*$, the module operates in the feed-limited regime (FLR). In this regime, the feed flow rate is lower than the distillate flow rate, and the feed temperature decreases along the module toward the distillate-side threshold temperature, $T_{\text{thres},l}$. This threshold is defined as the temperature at which the feed stream, with salinity C_s, has the same partial vapor pressure as the distillate stream, with $C_s = 0$, at the distillate inlet temperature $T_{e,l}$. Thus,

$$p_v\left(C_s, T_{\text{thres},l}\right) = p_v\left(0, T_{e,l}\right). \tag{86}$$

When the local temperature difference remains greater than the corresponding threshold temperature difference over the entire MD module, vapor transport occurs throughout the whole membrane length. Under these conditions, the process may become limited only by mass transfer, characterizing the mass-transfer-limited regime

(MTLR), which may result from insufficient membrane area or a very low membrane mass-transfer coefficient. Therefore, provided that mass-transfer limitations are negligible, a DCMD module operates under one of the two thermodynamically limited regimes: DLR or FLR.

According to Christie et al. [2020], for single-pass counter-current DCMD desalination, the FLR configuration is the most suitable operating condition for maximizing water recovery. Only under FLR conditions does the distillate stream have sufficient heat-capacity flow rate to sustain the maximum attainable evaporation of the feed stream. Consequently,

$$R_{\max}^{\mathrm{FLR}} > R_{\max}^{\mathrm{DLR}}. \tag{87}$$

Table 6 presents the average theoretical maximum water-recovery ratio, distillate mass flux, temperature-polarization coefficient, and thermal-performance indicators of the DCMD module, including the gain output ratio (GOR) and the specific thermal energy consumption, $\mathrm{SEC_{th}}$. Three scenarios involving the external heat exchangers were investigated for two feed-salinity levels and different feed and distillate flow rates: operation without heat recovery, with $\varepsilon_{\mathrm{HX},1} = \varepsilon_{\mathrm{HX},2} = 0$; operation with HX-1 only, with $\varepsilon_{\mathrm{HX},1} = 0.8$ and $\varepsilon_{\mathrm{HX},2} = 0$; and operation with both HX-1 and HX-2, with $\varepsilon_{\mathrm{HX},1} = \varepsilon_{\mathrm{HX},2} = 0.8$.

The results presented in Table 6 are consistent with the conclusions of Christie et al. [2020], indicating that the most favorable operating conditions correspond to FLR configurations in which $\alpha \approx \alpha^*$ and no threshold temperature is reached within the module. Under these conditions, high water-recovery ratios, RR, and the lowest values of $\mathrm{SEC_{th}}$ are obtained.

Under DLR conditions, for which $\alpha < \alpha^*$, decreasing the flow-rate ratio α reduces the water-recovery ratio and increases the specific thermal energy consumption. This behavior makes the configuration unattractive, even when the second heat exchanger, HX-2, is used to recover part of the thermal energy from the retentate stream. Conversely, under FLR conditions with $\alpha \gg \alpha^*$, the highest water-recovery ratios are achieved, although at the expense of substantially higher energy consumption.

Moreover, under these conditions, the inclusion of HX-2 produces no significant improvement in the overall performance of the DCMD system.

Table 6 also shows that, as the salinity of the brackish feed water increases, operating with a flow-rate ratio slightly greater than the critical value, α^*, can improve the water-recovery ratio, RR, the distillate volumetric flow rate, $\dot{V}_{\mathrm{jwl}}$, and the thermal efficiency, η_{th}, while causing only a small increase in $\mathrm{SEC}_{\mathrm{th}}$.

5.2.1 Operating Regime Analysis

After characterizing the DCMD desalination system, its behavior under different operating modes was investigated. Unless otherwise stated, the operating-mode simulations adopted the FLR configuration with $\dot{V}_l = \dot{V}_s = 50\ \mathrm{L\,min^{-1}}$. In addition, whenever no other condition is explicitly specified, both external heat exchangers, HX-1 and HX-2, were employed with effectiveness values of $\varepsilon_{\mathrm{HX},1} = 0.8$ and $\varepsilon_{\mathrm{HX},2} = 0.8$, respectively.

To assess whether the external heat exchangers improve the energy efficiency of the system, Figures 8 and 9 illustrate the dynamic behavior of GOR and $\mathrm{SEC}_{\mathrm{th}}$ under batch operation for two maximum allowable salt concentrations in the mixing tank: $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$ and $C_{s,\max} = 260\ \mathrm{g_s\,L_{mix}^{-1}}$, respectively. The average values shown in these figures, $\overline{\mathrm{GOR}}$ and $\overline{\mathrm{SEC}}_{\mathrm{th}}$, were calculated over an 8 h operating period.

Table 6: Influence of the operational conditions on the averaged DCMD module efficiency metrics under steady-state operation over 8 h.

Flow condition [L/min]	$\boldsymbol{\alpha}$	$\boldsymbol{\alpha^*}$	Oper.	$\boldsymbol{T_{\mathrm{thr}}}$ [°C]	$\boldsymbol{Z_{\mathrm{thr}}}$ [m]	$\boldsymbol{RR}$	$\overline{\dot{\boldsymbol{V}}}_{\mathbf{jwl}}$ [L min^{-1}]	$\overline{\mathbf{TPC}}$ [−]	$\overline{\boldsymbol{\eta}}_{\mathbf{th}}$ [%]	$\overline{\mathbf{GOR}}$ [−]			$\overline{\mathbf{SEC}}_{\mathbf{th}}$ [kWh m^{-3}]		
										$\varepsilon_{\mathrm{HX},1}=0.0$ $\varepsilon_{\mathrm{HX},2}=0.0$	$\varepsilon_{\mathrm{HX},1}=0.8$ $\varepsilon_{\mathrm{HX},2}=0.0$	$\varepsilon_{\mathrm{HX},1}=0.8$ $\varepsilon_{\mathrm{HX},2}=0.8$	$\varepsilon_{\mathrm{HX},1}=0.0$ $\varepsilon_{\mathrm{HX},2}=0.0$	$\varepsilon_{\mathrm{HX},1}=0.8$ $\varepsilon_{\mathrm{HX},2}=0.0$	$\varepsilon_{\mathrm{HX},1}=0.8$ $\varepsilon_{\mathrm{HX},2}=0.8$
$w_{\mathrm{NaCl},s}=3.5\%\approx35$ g$_{\mathrm{s}}$ L$^{-1}_{\mathrm{mix}}$															
$\dot{V}_l=30<\dot{V}_s=80$	0.38	0.88	DLR	80.0[a]	0.63	2.5	2.6	0.73	73	0.28	0.41	1.4	2500	1700	480
$\dot{V}_l=30<\dot{V}_s=50$	0.60	0.88	DLR	80.0[a]	0.98	3.7	2.5	0.62	69	0.43	0.90	2.4	1600	780	290
$\dot{V}_l=50<\dot{V}_s=80$	0.63	0.88	DLR	—	—	4.0	4.2	0.50	71	0.45	0.97	2.5	1600	720	280
$\dot{V}_l\approx44<\dot{V}_s=50$	0.88	0.88	FLR	—	—	5.0	3.3	0.56	67	0.56	1.9	3.3	1200	360	210
$\dot{V}_l=\dot{V}_s=30$	1.0	0.88	FLR	—	—	5.3	2.1	0.61	54	0.60	2.1	2.6	1200	330	270
$\dot{V}_l=\dot{V}_s=50$	1.0	0.88	FLR	—	—	5.2	3.4	0.59	66	0.58	1.8	2.5	1200	380	270
$\dot{V}_l=\dot{V}_s=80$	1.0	0.88	FLR	—	—	4.9	5.2	0.58	67	0.55	1.5	2.4	1300	460	290
$\dot{V}_l=80>\dot{V}_s=50$	1.6	0.88	FLR	—	—	5.3	3.5	0.66	62	0.60	1.1	1.2	1200	620	580
$\dot{V}_l=50>\dot{V}_s=30$	1.7	0.88	FLR	—	—	5.5	2.2	0.69	62	0.62	1.1	1.2	1100	600	590
$\dot{V}_l=80>\dot{V}_s=30$	2.7	0.88	FLR	25.3[b]	0.033	5.6	2.2	0.72	61	0.63	0.89	0.89	1100	780	780
$w_{\mathrm{NaCl},s}=12.0\%\approx260$ g$_{\mathrm{s}}$ L$^{-1}_{\mathrm{mix}}$															
$\dot{V}_l=30<\dot{V}_s=80$	0.33	0.74	DLR	79.6[a]	0.41	0.57	0.69	0.87	20	0.076	0.11	0.39	8900	6200	1800
$\dot{V}_l=30<\dot{V}_s=50$	0.53	0.74	DLR	79.0[a]	0.63	0.92	0.70	0.83	20	0.12	0.24	0.63	5500	2800	1100
$\dot{V}_l=50<\dot{V}_s=80$	0.55	0.74	DLR	—	—	1.9	2.3	0.73	42	0.26	0.54	1.4	2700	1300	500
$\dot{V}_l\approx42<\dot{V}_s=50$	0.74	0.74	FLR	—	—	2.2	1.7	0.72	37	0.30	0.88	1.6	2300	790	430
$\dot{V}_l=\dot{V}_s=30$	0.88	0.74	FLR	—	—	1.9	0.87	0.75	29	0.26	0.78	0.99	2700	890	710
$\dot{V}_l=\dot{V}_s=50$	0.88	0.74	FLR	—	—	2.6	2.0	0.71	42	0.36	0.98	1.4	1900	710	510
$\dot{V}_l=\dot{V}_s=80$	0.88	0.74	FLR	—	—	2.9	3.6	0.68	50	0.40	0.96	1.5	1700	720	460
$\dot{V}_l=80>\dot{V}_s=50$	1.4	0.74	FLR	—	—	3.1	2.3	0.74	45	0.42	0.74	0.81	1600	940	860
$\dot{V}_l=50>\dot{V}_s=30$	1.5	0.74	FLR	30.9[b]	0.44	2.8	1.3	0.80	39	0.38	0.66	0.69	1800	1100	1000
$\dot{V}_l=80>\dot{V}_s=30$	2.3	0.74	FLR	29.8[b]	0.55	3.0	1.4	0.83	40	0.41	0.56	0.57	1700	1300	1200

[a] Feed-side threshold temperature, $T_{\mathrm{thres},s}$. [b] Distillate-side threshold temperature, $T_{\mathrm{thres},l}$.

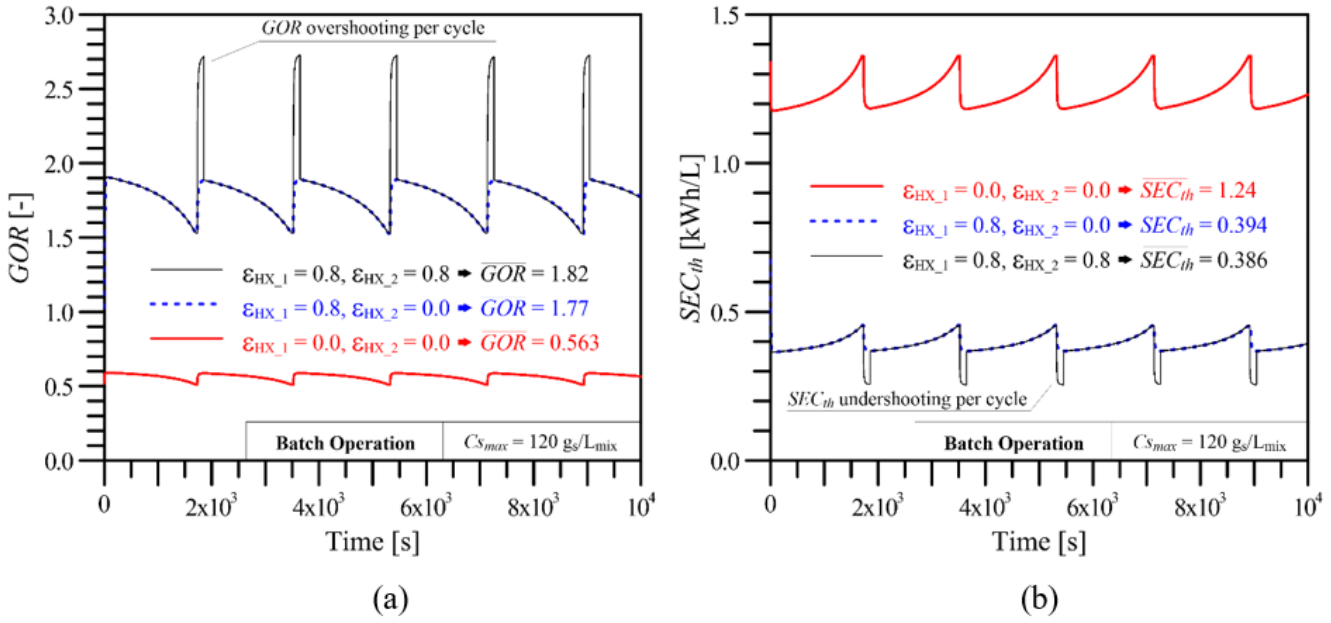


Figure 8: Evolution of the instantaneous (a) GOR and (b) $\mathrm{SEC_{th}}$ for $C_{s,\mathrm{max}} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$ under batch operation.

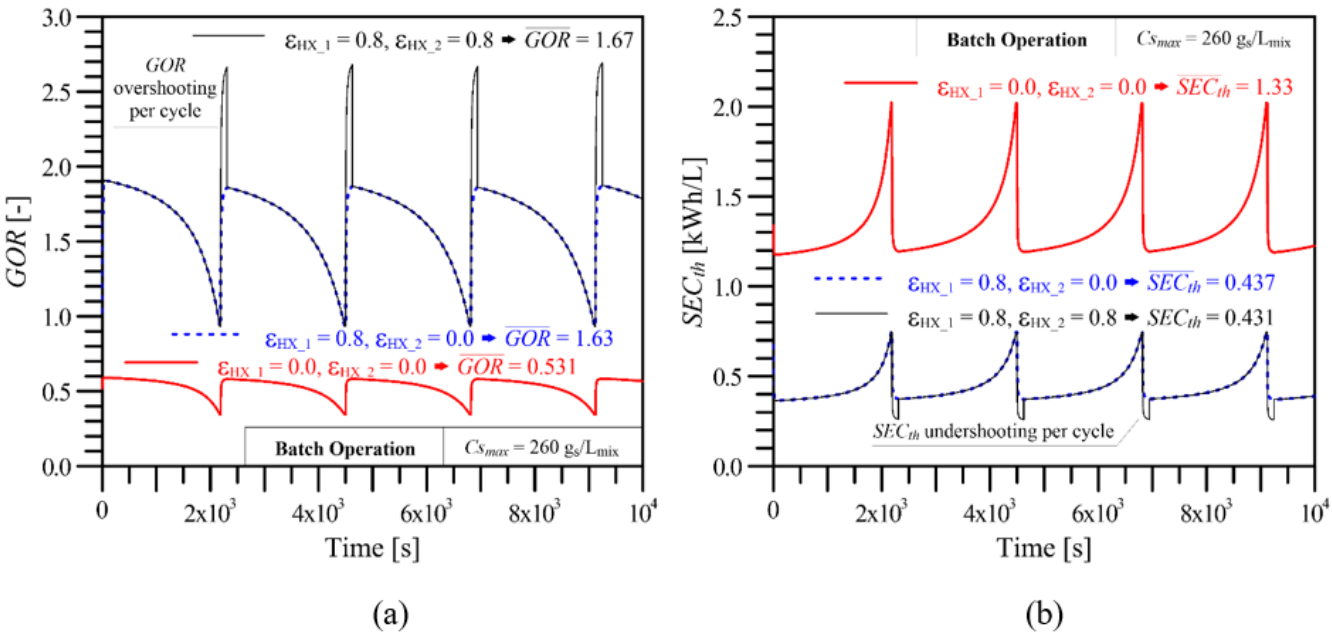


Figure 9: Evolution of the instantaneous (a) GOR and (b) $\mathrm{SEC_{th}}$ for $C_{s,\mathrm{max}} = 260\ \mathrm{g_s\,L_{mix}^{-1}}$ under batch operation.

Figures 8 and 9 show that the improvement in thermal efficiency obtained by using HX-2, represented by an increase in $\overline{\mathrm{GOR}}$ and a reduction in $\overline{\mathrm{SEC_{th}}}$, is associated with the overshoot in the instantaneous GOR and the undershoot in the

instantaneous $\mathrm{SEC_{th}}$. These effects occur while the mixing tank is being emptied and refilled during each cycle of the batch operating mode, as well as in the other cyclic operating modes.

This behavior can also be observed in Figure 10, which presents the temporal evolution of the main DCMD-system parameters under the batch, semi-batch, volume-control (V-Control), continuous, and steady-state operating modes. The evaluated parameters are the salt concentration in the mixing tank, $C_{s,\mathrm{mix}}$; the volumetric flow rate of the produced distillate, $\dot{V}_{\mathrm{jwl}}$; the mixture volume in the mixing tank, V_{mix}; the mixture temperature, T_{mix}; the rejection fraction, λ; and the volumetric flow rate supplied by the brackish-water well, $\dot{V}_{\mathrm{well}}$.

Several characteristics of the operating modes can be identified from these results. Under batch operation, the mixture salinity varies between the brackish-water well salinity, $C_{s,\mathrm{well}} \approx 35\ \mathrm{g_s\,L_{mix}^{-1}}$, and the specified maximum salinity, $C_{s,\mathrm{max}} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$, during each cycle as distillate is produced, as shown in Figure 10(a). A similar variation is observed for the produced-distillate flow rate, $\dot{V}_{\mathrm{jwl}}$, which decreases from approximately 2.596 to 2.256 $\mathrm{L\,min^{-1}}$ as the mixture salinity increases, as shown in Figure 10(b).

The semi-batch mode exhibits similar behavior, although with a considerably lower cycling frequency than the batch mode. In the V-Control mode, the salinity and distillate flow rate also oscillate between the aforementioned limits. However, intermediate oscillations occur within each main cycle because the mixing-tank volume is constrained between the prescribed minimum and maximum values of 60 and 140 L, respectively.

The cycle duration of each transient operating mode can be identified by the sharp decrease in the mixing-tank volume shown in Figure 10(c), which occurs when the solution is discharged from the tank, or by the abrupt increase in the rejection fraction shown in Figure 10(e). These events are accompanied by corresponding variations in the mixture temperature, Figure 10(d), and in the brackish-water flow rate, Figure 10(f). The cycle duration is approximately 1789 s ($\approx$ 30 min) for batch operation, 4800 s ($\approx$ 1 h 20 min) for V-Control operation, and 6090 s ($\approx$ 1 h 41 min) for semi-batch operation.

Although the system is emptied and subsequently refilled with fresh brackish

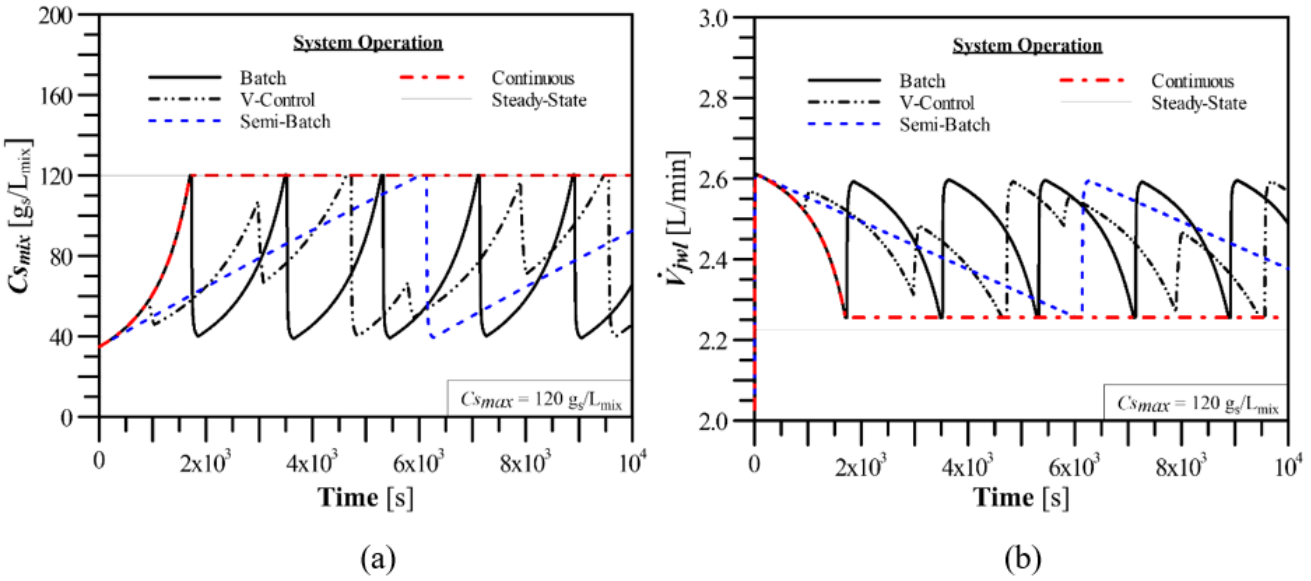


Figure 10: Dynamic evolution of the instantaneous desalination-system parameters under different operating modes for $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$: (a) $C_{s,\mathrm{mix}}$ and (b) $\dot{V}_{\mathrm{jwl}}$.

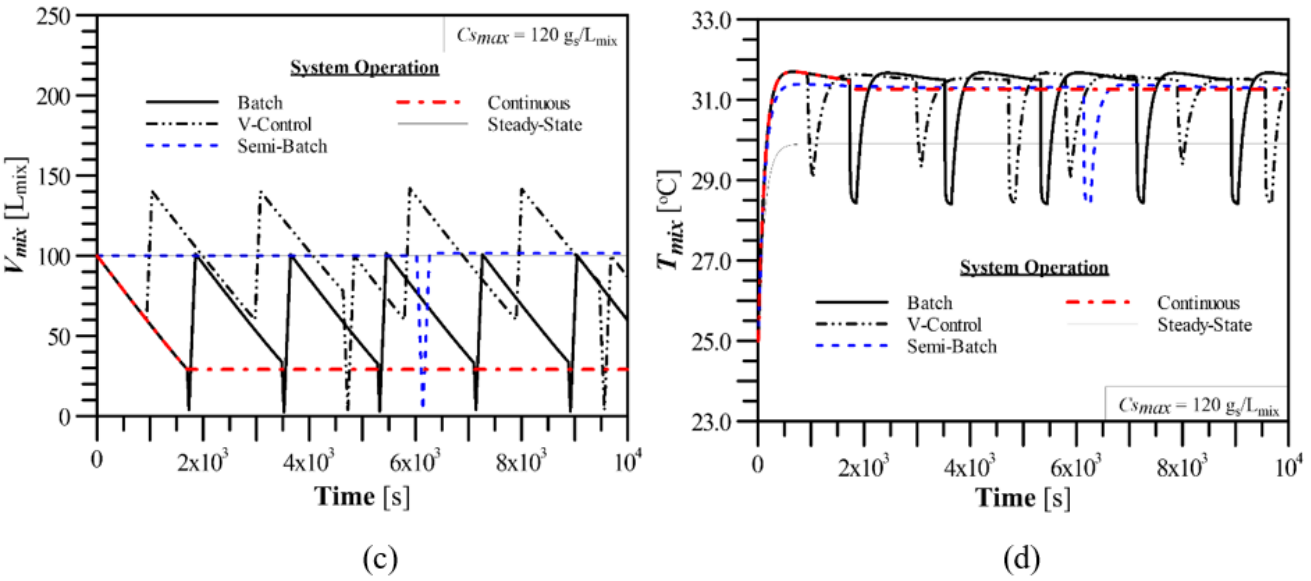


Figure 10: Dynamic evolution of the instantaneous desalination-system parameters under different operating modes for $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$, continued: (c) V_{mix} and (d) T_{mix}.

water from the well at $T_{\mathrm{well}} = 25\ ^{\circ}\mathrm{C}$, the temperature of the mixture in the tank does not return to this value during each cycle because of the thermal inertia of the system. In the procedure adopted here, emptying and refilling occur sequentially: the tank is continuously emptied until a predefined minimum volume is reached and is then continuously refilled to the initial volume, $V_{\mathrm{mix},0} = 100$ L.

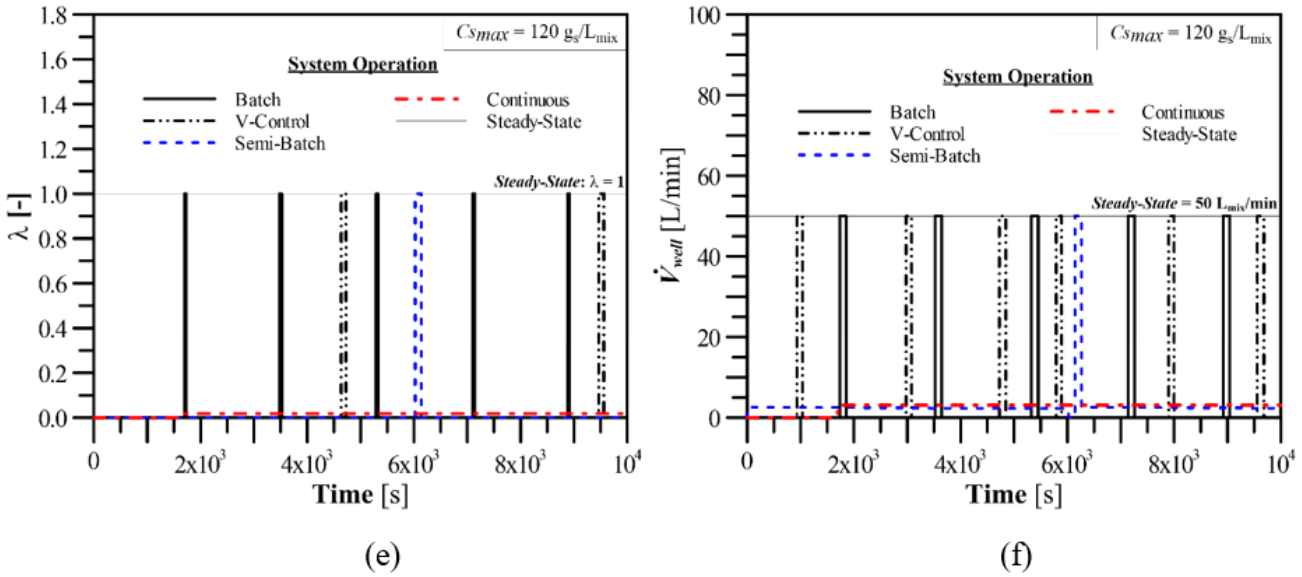


Figure 10: Dynamic evolution of the instantaneous desalination-system parameters under different operating modes for $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$, continued: (e) λ and (f) $\dot{V}_{\mathrm{well}}$.

To identify the most appropriate operating mode from the perspectives of energy efficiency and water recovery, Figures 11 and 12 compare the average specific thermal energy consumption, $\overline{\mathrm{SEC}}_{\mathrm{th}}$, and the water recovery ratio, RR, respectively.

The black bars with gradient and solid fills represent the scenarios with maximum salt concentrations of $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$ and $C_{s,\max} = 260\ \mathrm{g_s\,L_{mix}^{-1}}$, respectively. Thus, under the transient batch, V-Control, and semi-batch modes, the salinity in the mixing tank varies from the brackish-water well salinity to the specified value of $C_{s,\max}$. In the steady-state modes, operation at a prescribed value of $C_{s,\max}$ requires the final brine or the brackish-water/brine stream supplied to the system to have the same salinity; consequently, $C_{s,\mathrm{oper}} = C_{s,\max}$.

The blue bars shown for the continuous/steady-state cases represent scenarios in which $C_{s,\mathrm{oper}} = C_{s,\max}$, with salinities of 66 and 97 $\mathrm{g_s\,L_{mix}^{-1}}$. These values correspond to the average system salinities under batch operation. In steady-state operation, because the brine leaving the DCMD module is continuously discharged, a saline stream with a prescribed concentration is continuously supplied to the system at the predefined flow rate of 50 $\mathrm{L\,min^{-1}}$. The concentrations represented by the green bars are 66 and 97 $\mathrm{g_s\,L_{mix}^{-1}}$, whereas those represented by the black bars are 120 and

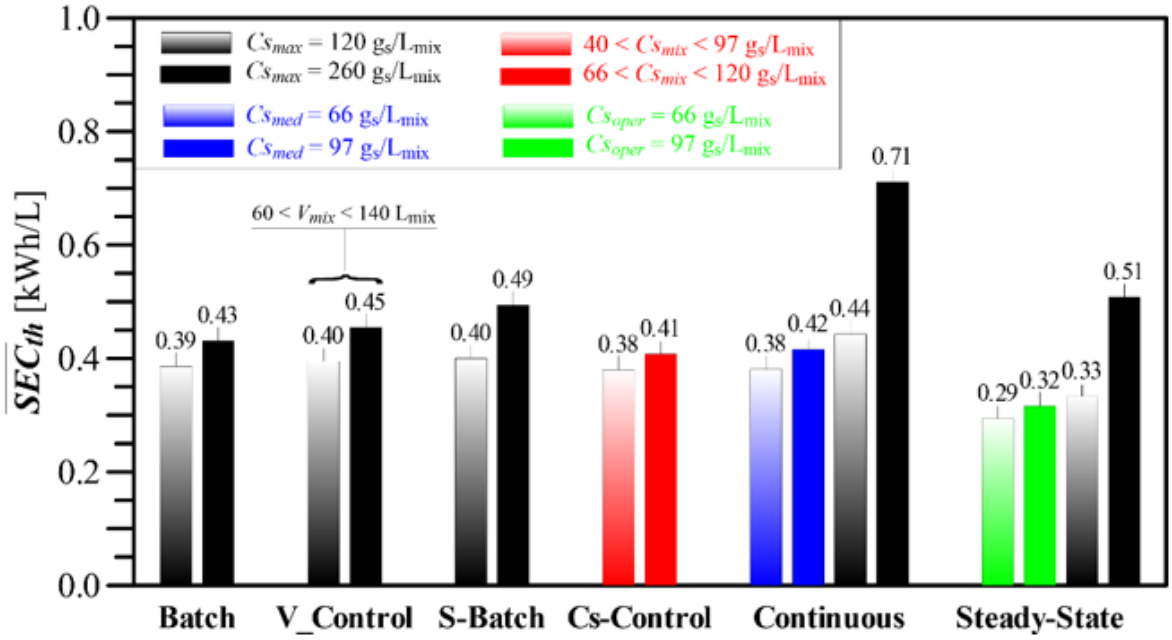


Figure 11: Effect of the system operating mode on the average specific thermal energy consumption, $\overline{\mathrm{SEC}}_{\mathrm{th}}$, for different mixture salinities.

260 $g_s\,L_{mix}^{-1}$.

The red bars represent two salinity-control scenarios in which the concentration of the solution in the mixing tank is constrained to vary from 40 to 97 $g_s\,L_{mix}^{-1}$ and from 66 to 120 $g_s\,L_{mix}^{-1}$ for the gradient-filled and solid-filled bars, respectively.

Overall, these results show that a balance must be sought between specific thermal energy consumption and water recovery because, for a given maximum operating salinity, a trade-off exists between these metrics. In other words, low specific thermal energy consumption and high water recovery cannot generally be achieved simultaneously.

For the same limiting mixture salinity, batch operation provides the lowest energy consumption, followed by the semi-batch, steady-state, and continuous modes, in agreement with the literature [Swaminathan and Lienhard, 2018]. Regarding water recovery, Figure 12 shows that the semi-batch mode provides the highest recovery ratio, followed by the V-Control and batch modes. Steady-state operation is unfavorable from this perspective unless the system is of the type studied by Sampaio [2022], for which complete rejection of the retentate leaving the desalination module is acceptable.

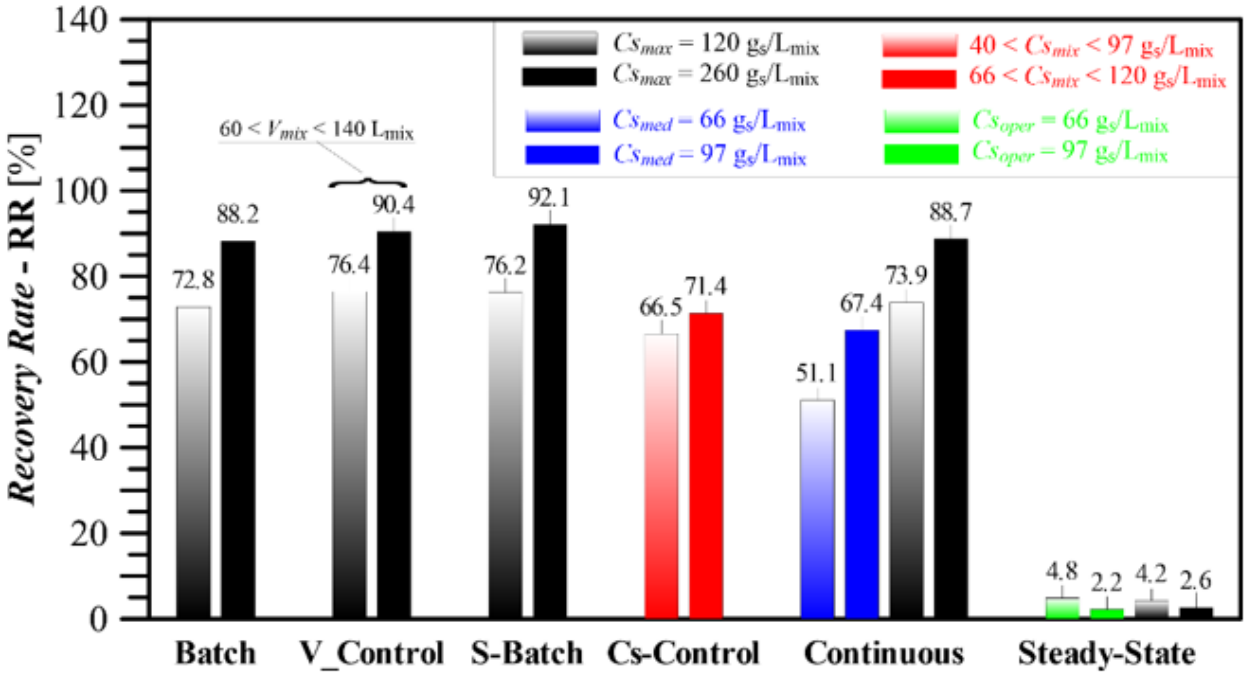


Figure 12: Effect of the system operating mode on the water recovery ratio, RR, for different mixture salinities.

The performance of the V-Control mode at an intermediate salt concentration is particularly noteworthy. For $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$, this mode produces slightly more distillate per unit volume of brackish well water than the semi-batch mode, with recovery values of 76.4 $\mathrm{L_{dist}\,L_{well}^{-1}}$ and 76.2 $\mathrm{L_{dist}\,L_{well}^{-1}}$, respectively, at the same average specific thermal energy consumption, $\overline{\mathrm{SEC}}_{\mathrm{th}} = 0.40\ \mathrm{kWh\,L^{-1}}$. However, this difference may be negligible because the energy requirements associated with process control and intermittent pumping were not included in the present analysis.

Figure 12 also shows that operating the desalination unit continuously at a salinity equal to the average salinity of the batch mode, represented by the blue bars, can result in lower energy consumption than batch operation, but at the expense of a lower water recovery ratio.

Figure 13 compares the volume of distillate produced by each operating mode over an 8 h period.

At the lower salinity level, the desalination system can produce up to approximately 1,200 L of distillate over 8 h under any of the evaluated operating modes. At the higher salinity level, distillate production decreases, particularly under the

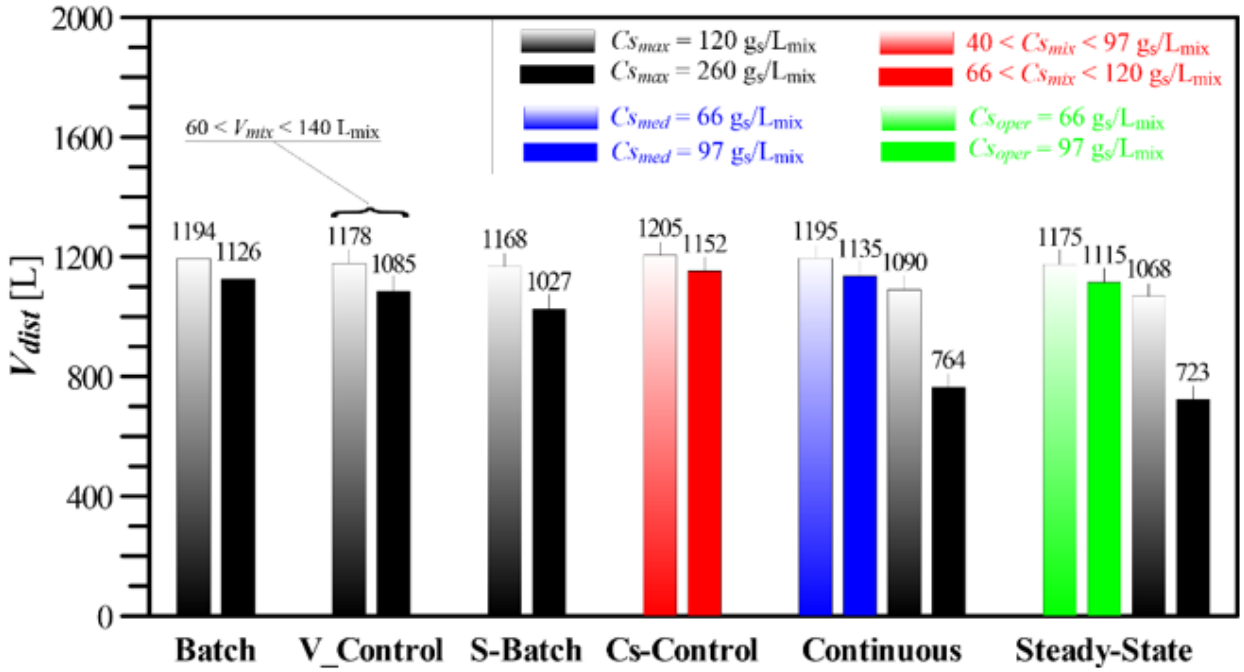


Figure 13: Effect of the system operating mode on the volume of distillate produced over 8 h for different mixture salinities.

continuous and steady-state modes, for which only approximately 750 L is produced.

Therefore, for implementation in regions subject to seasonal water stress, such as the semi-arid region of northeastern Brazil, the results indicate that semi-batch operation may be the preferred alternative for brackish-water desalination when the primary objective is to minimize the consumption of brackish well water. Nevertheless, the energy costs associated with pumps, control devices, and other accessories required by each operating mode should be included in a broader thermoeconomic analysis.

6 Conclusions

The present work developed a transient physical-mathematical model and its computational implementation for a hollow-fiber DCMD desalination plant operating with dual heat recovery from the distillate and retentate streams and with total or partial recirculation of the concentrated stream. Six operating modes were examined under conditions representative of brackish-water wells in the semi-arid Brazilian northeast-

ern region, allowing energy efficiency and water conservation to be assessed together rather than in isolation.

Beyond confirming that heat recovery from the distillate is a mandatory feature of any competitive DCMD plant, the results reposition the retentate stream, which conventional designs discard together with the concentrated brine, as a thermal resource whose recovery through a second heat exchanger substantially alters the standing of DCMD in the desalination landscape. The magnitude of the efficiency gain enabled by HX-2 is large enough to narrow the historical gap between thermal separation and reverse osmosis in the small-to-medium capacity range (< 100 m^3/day), which is precisely the range in which RO loses competitiveness because of high specific capital cost and sensitivity to feed variability. This is a qualitative shift more than a numerical one: it indicates that DCMD, when equipped with dual heat recovery, is not merely a niche technology for waste-heat valorization but a viable alternative wherever low-grade thermal energy (solar collectors, industrial residual heat, engine exhaust) is locally available and RO scaling is uneconomical.

A second consequence of the transient analysis concerns the role of the critical mass-flow-rate ratio, α^*, as an *operational* rather than merely *design* parameter. Because the temperature and salinity of the well feed vary seasonally in hydrologically stressed regions, and because α^* depends on both, fixed-flow DCMD operation is inherently suboptimal for such applications: the proximity to α^* determines not only the water-recovery ratio but also whether HX-2 delivers a meaningful benefit at all. Under FLR conditions with $\alpha \gg \alpha^*$, the retentate leaves the module already close to ambient temperature and little sensible heat remains to be recovered, so the second heat exchanger becomes essentially inert. This has direct implications for control-system design: adaptive strategies that track α^* as the feed drifts would extract considerably more value from the dual-recovery architecture than the fixed operating points typically reported in the literature.

The comparison across operating modes reveals a trade-off that cannot be resolved by pure thermodynamic optimization. Batch operation minimizes specific thermal energy consumption because the tank's thermal inertia keeps the average feed salinity, and thus the transmembrane vapor-pressure driving force, favorable during most of each cycle, but it periodically discharges brackish water that has

not been fully exploited. Semi-batch operation eliminates this waste and achieves recovery ratios above ninety percent, at the cost of a slightly higher energy demand, because fresh well water is supplied at the same rate at which distillate is produced and no volume is lost during flush-and-refill transients. For the semi-arid Brazilian context that motivated this study, and for analogous regions in which the shadow price of well water exceeds the shadow price of thermal energy, a common situation whenever solar collectors or industrial waste heat are locally available, semi-batch operation is therefore the physically appropriate choice, not the mode that a purely energy-based figure of merit would select. The volume-controlled mode emerges as a practical middle ground that retains most of the semi-batch water recovery while preserving batch-level energy efficiency, and deserves attention in future pilot deployments.

More broadly, the transient framework developed here calls into question the widespread practice of characterizing MD desalination plants through steady-state metrics alone. The overshoots in GOR and undershoots in $\mathrm{SEC_{th}}$ observed during each flush-and-refill are not artifacts to be averaged away; they are the physical mechanism through which HX-2 delivers most of its benefit, because it is precisely during those transients that hot retentate flows past HX-2 while cold well water is being drawn in. Any economic assessment or control-loop tuning built on steady-state assumptions will therefore systematically underestimate the value of dual heat recovery and, by extension, the competitiveness of DCMD for the small-scale applications discussed above.

Two lines of extension follow naturally. The first is the incorporation of the pumping, control, and instrumentation costs that were deliberately excluded from the present energy-efficiency analysis, so that a full thermoeconomic ranking of the six operating modes can be produced, a step required before any recommendation on operating mode can be turned into a design specification for a real plant. The second is the extension of the modeling framework to AGMD and V-AGMD configurations, which offer higher intrinsic thermal efficiencies but were not addressed here, in order to simulate the entire TRL-6 experimental desalination unit installed at LabMEMS/UFRJ, which is controlled through the volume of the collector tank. Together, these developments should support the transition of hollow-fiber MD from

a laboratory technique toward field deployment in the specific regional context that motivated this study.

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